

Module 4

Finite State Machines

Finite State Machines

- Finite State Machines (FSMs) are at the heart of most digital design.
- The basic idea of an FSM is to store a sequence of different unique states and transition between them depending on the values of the inputs and the current state of the machine.
- Finite state machine is a graphical model/representation of sequential activities or events.
- After representing and modelling the events they can be implemented easily in case of sequential logic designs.

Applications of FSM

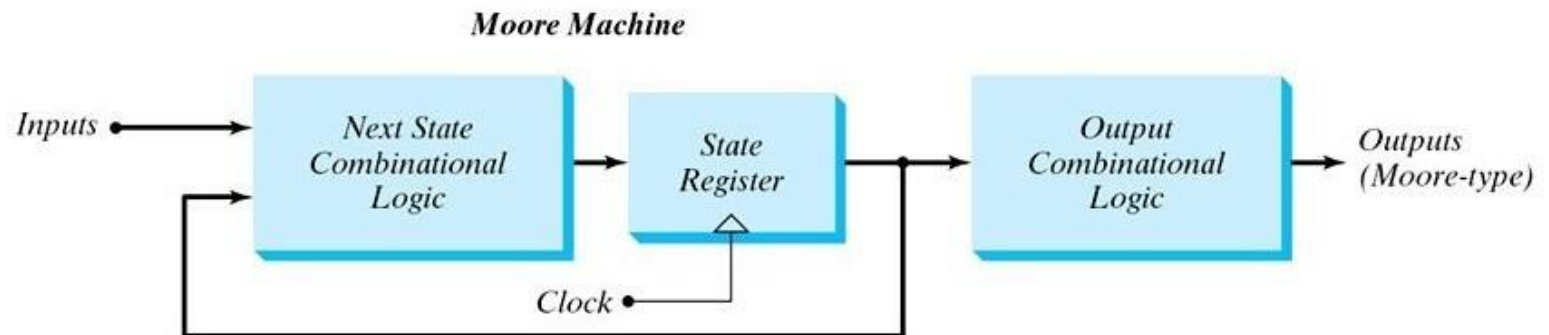
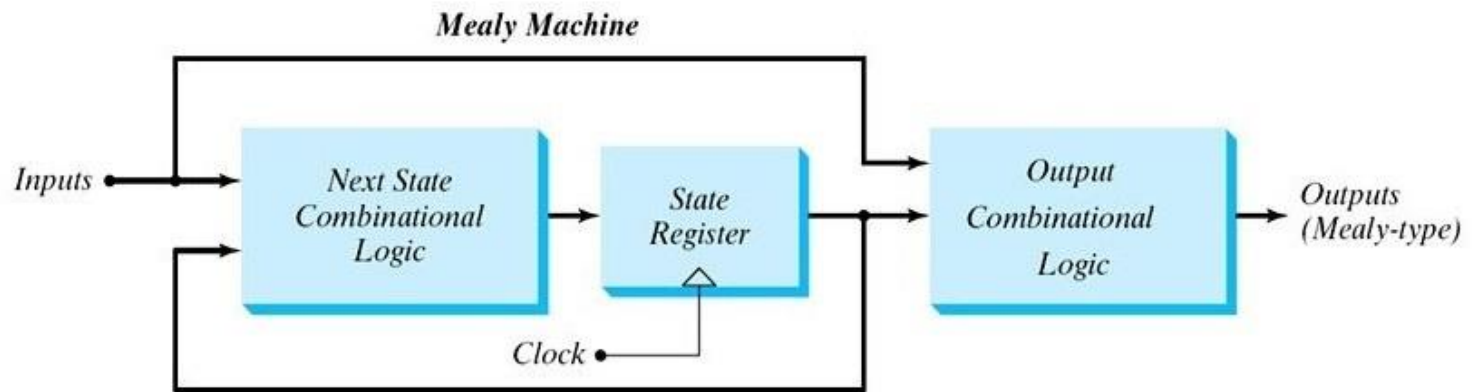
Finite state machines can be utilized in many fields of study:

- neural networks,
- artificial intelligence,
- mathematics,
- games,
- robotics
- sequential flow of data.
- in a station the vending machine which dispatches ticket uses a simple FSM.
- In the complex digital systems the controlling part is most of the times implemented using FSMs

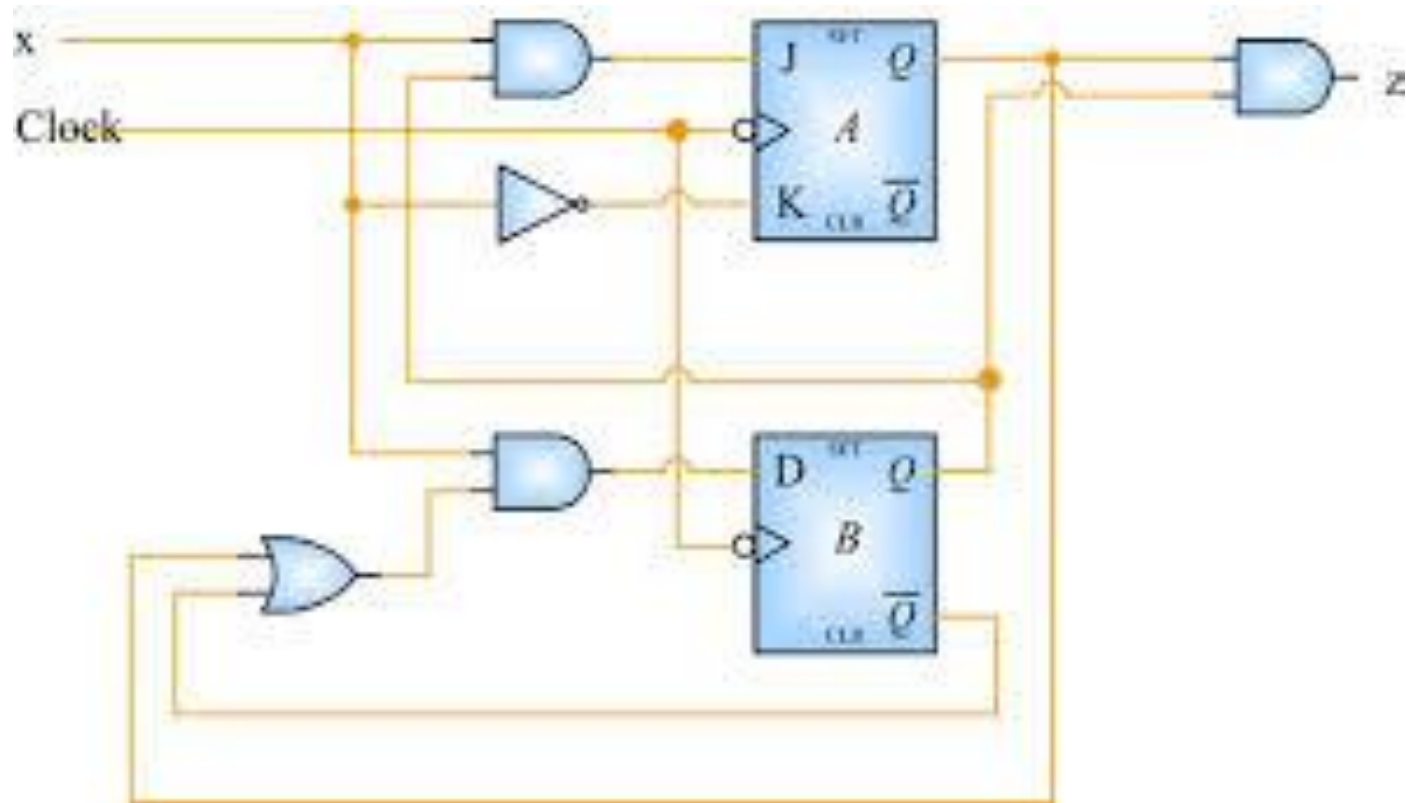
Types of FSM

- Synchronous sequential circuits are represented by two models:
 - **Moore Model**- The output depends only on the present state of the flip-flops
 - **Mealy Model**- The output depends on both the present state of the flip flop(s) and on the input(s)

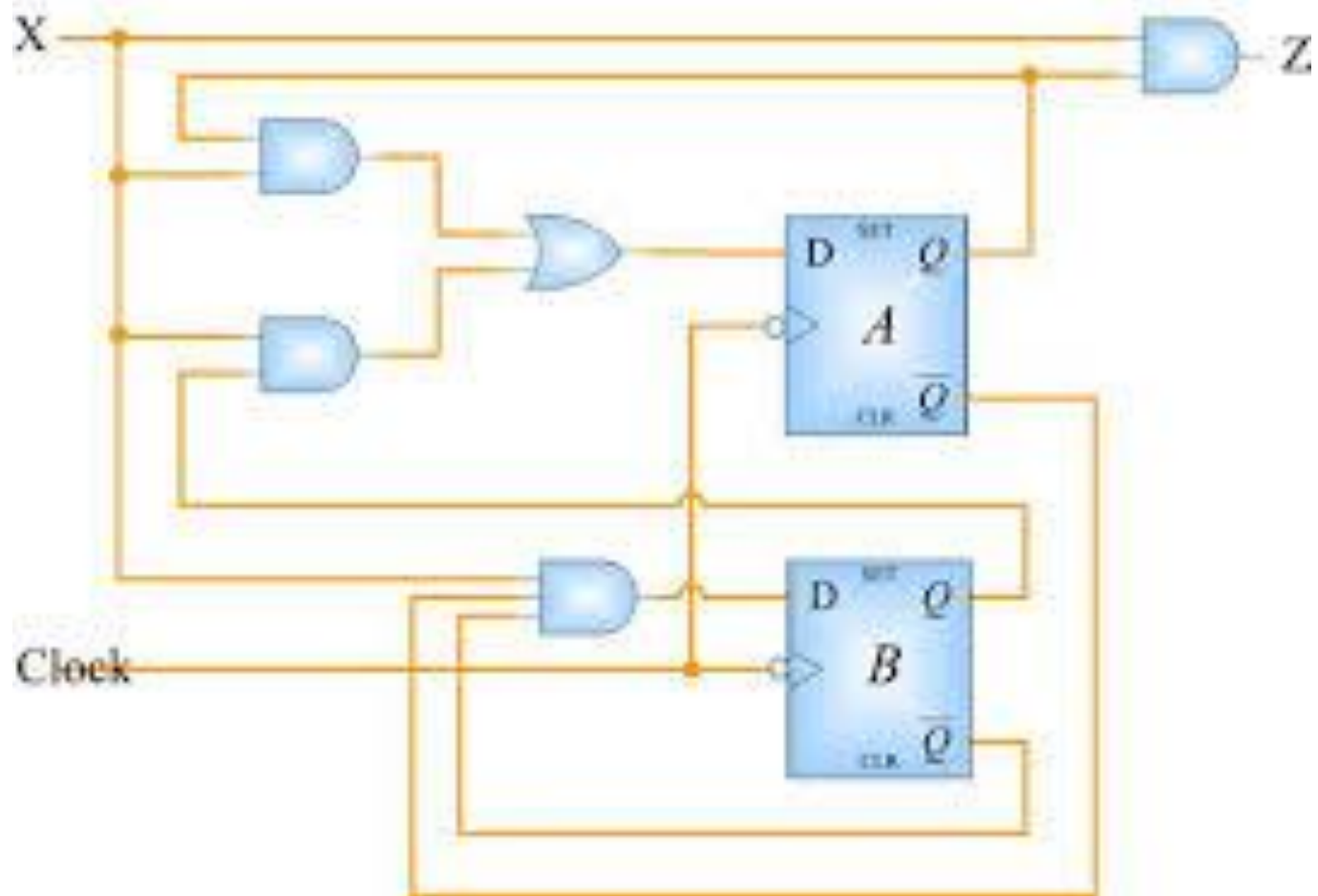
Block diagram of Moore and Mealy models



Example of Moore machine

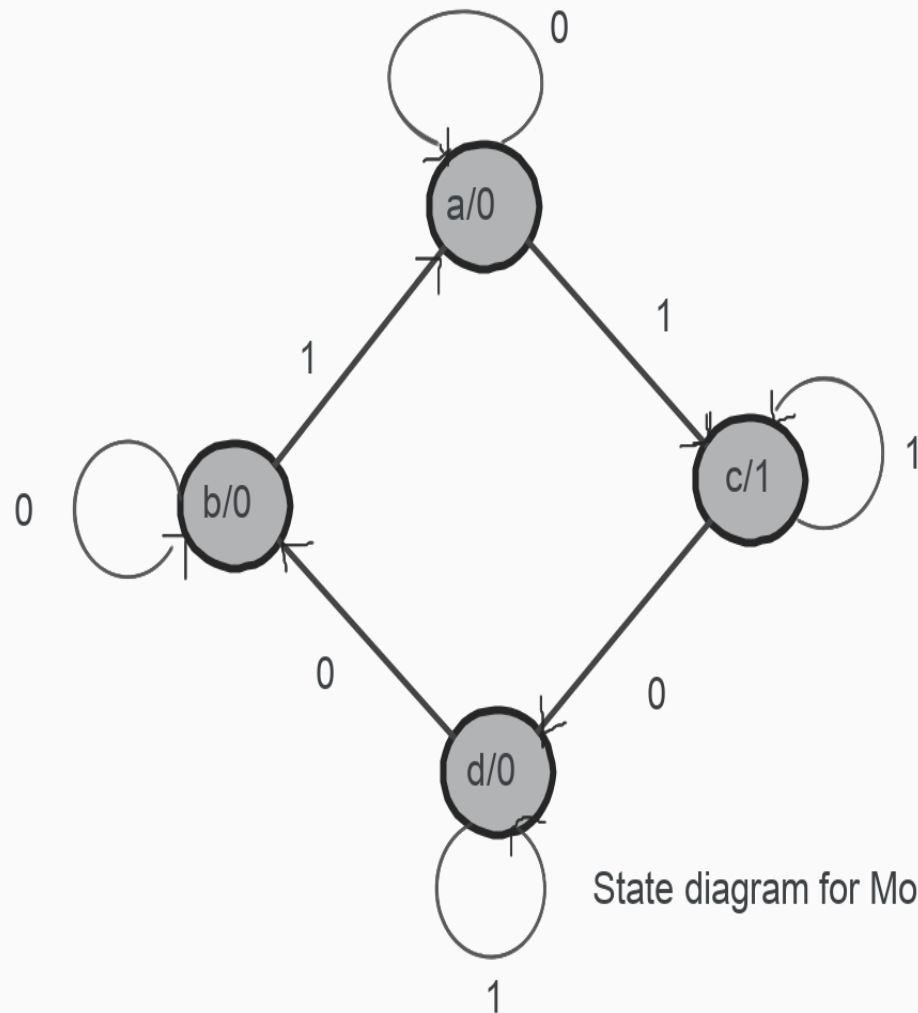


Example of Mealy machine



State Transition/State diagram

- Pictorial representation of a behaviour of a sequential circuit



State diagram for Moore circuit

State diagram of Moore circuit

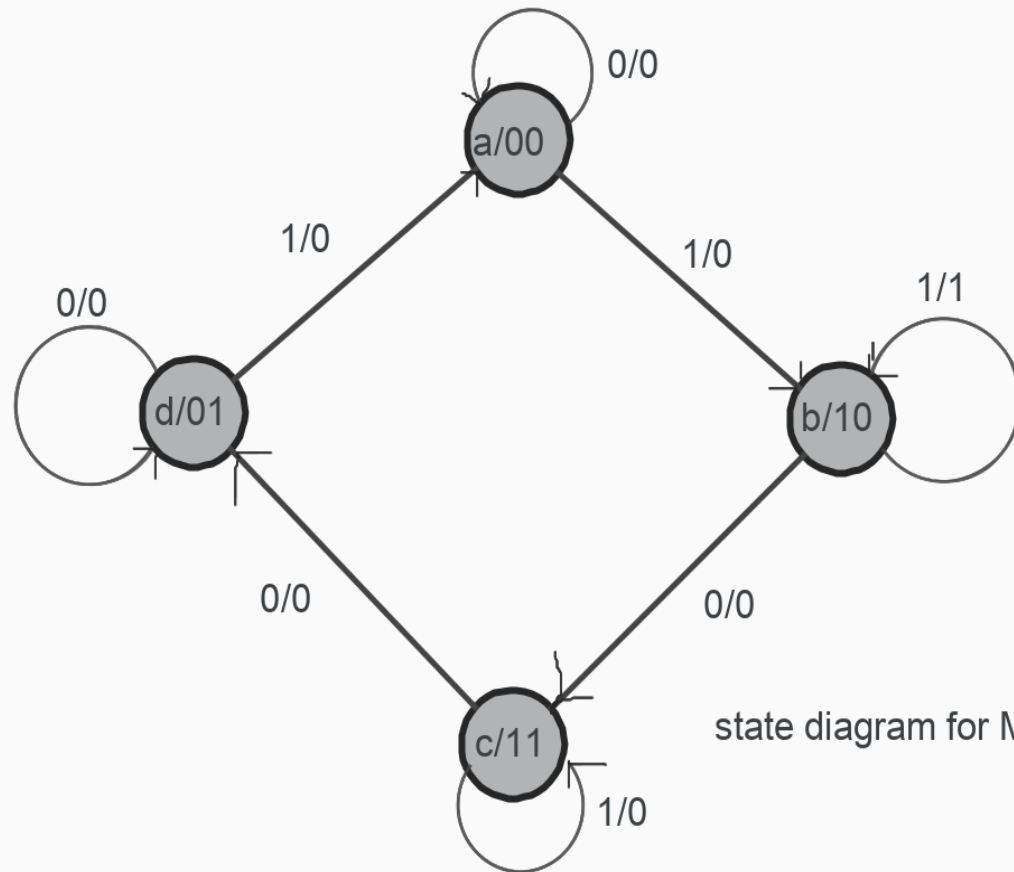
- The state is represented by the circle
- Transition between states is indicated by directed lines connecting circles
- A directed line connecting a circle with itself indicates that next state is same as present state
- Binary number inside each identifies the state represented by the circle
- In case of Moore circuit, the directed lines are labelled with only one binary number representing the state of the input that causes the state transition
- The output state is indicated within the circle, below the present state because output depends only on the present state

State Table :Moore circuit

- Information contained in the state diagram is translated into a tabular form.

Present state	Next state		Output
	X=0	X=1	
AB	AB	AB	Y
a	a	c	
b	b	a	0
c	d	c	1
d	b	d	0

State diagram of Mealy circuit



state diagram for Mealy circuit

State diagram of Mealy circuit

- The state is represented by the circle
- Transition between states is indicated by directed lines connecting circles
- A directed line connecting a circle with itself indicates that next state is same as present state
- Binary number inside each identifies the state represented by the circle
- The directed lines are labelled with two binary numbers separated by a symbol “ / “
- Input value that causes the state transition is labelled first and the output value during the present state is labelled after the symbol “/ ” for Mealy circuits

State Table :Mealy circuit

Present state	Next state		Output	
	X=0	X=1	X=0	X=1
AB	AB	AB	Y	Y
a	a	c	0	0
b	b	a	0	0
c	d	c	0	1
d	b	d	0	0

Transition Table

- A transition table takes the state table one step further
- In transition table, specific state variable values are assigned to each state
- Assignment of values to state variables is called State Assignment
- Transition table also represents relationship between input, output and flip-flop states

Transition Table

Present state		Next state		Output	
		X=0	X=1	X=0	X=1
A	B	AB	AB	Y	Y
0	0	00	10	0	1
0	1	11	00	0	0
1	0	10	01	1	0
1	1	00	10	1	0

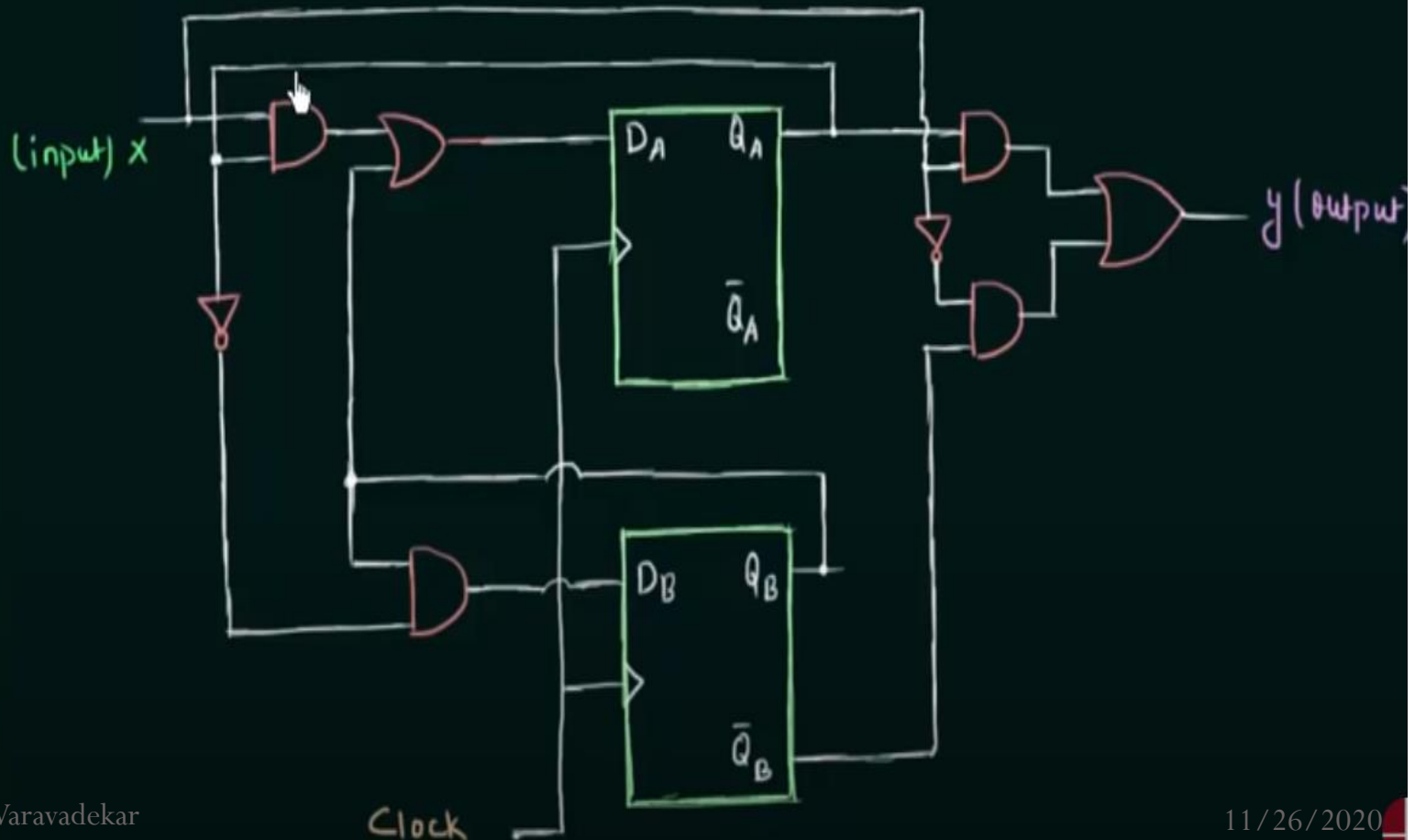
Analysis of clocked Sequential circuits

- Step 1: Determine FF input equations and the output equations from the given sequential circuit
- Step 2: Derive the transition equation
- Step 3: Plot the next step map for each FF
- Step 4: Plot the transition table
- Step 5: Draw the state table
- Step 6: Draw the state diagram

Example1

Analysis of Clocked Sequential Circuits (with D Flip Flop)

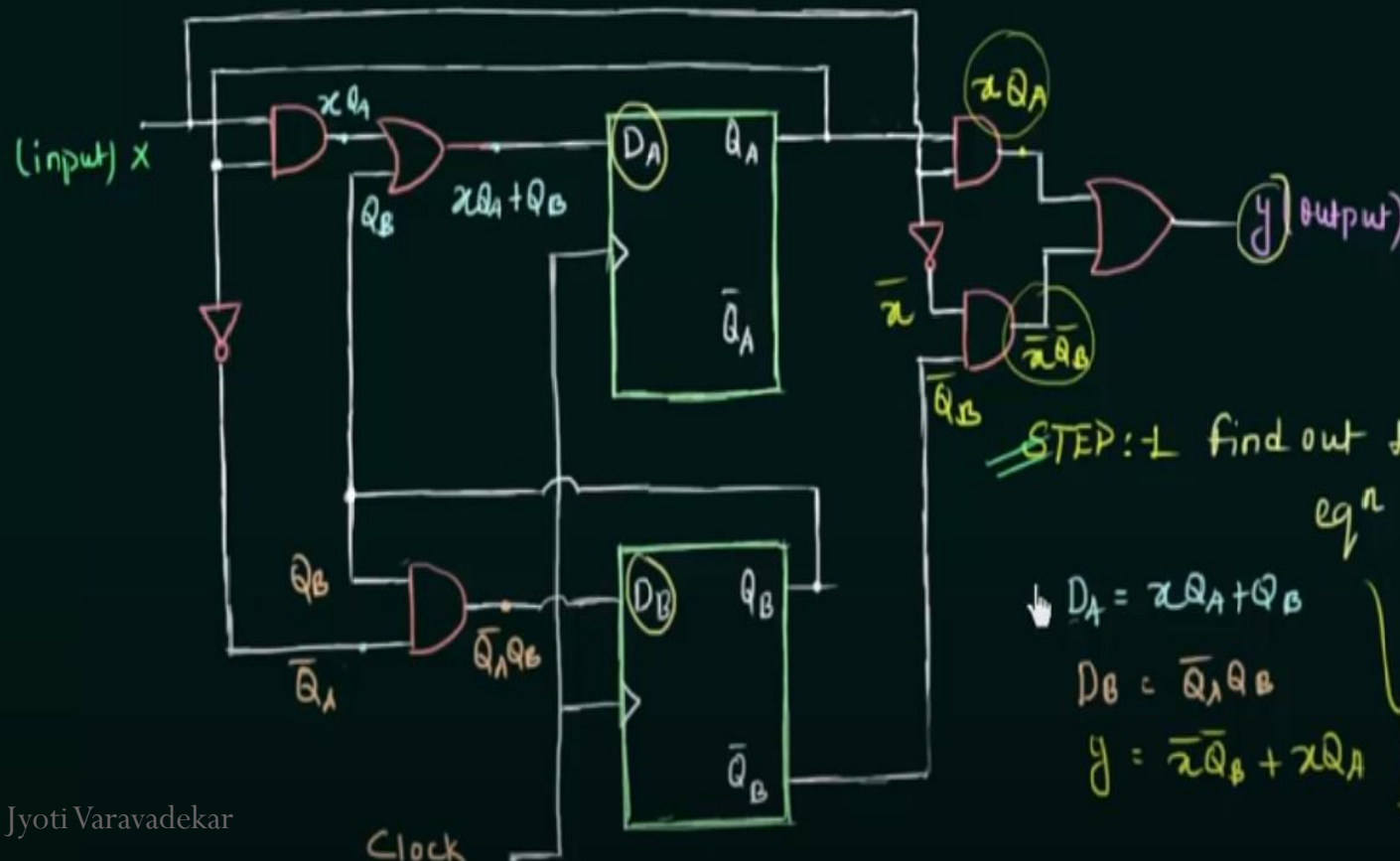
Analysis of Clocked Sequential Circuits (with D Flip Flop)



Example 1: Step 1: I/P and O/P Equations

Analysis of Clocked Sequential Circuits (with D Flip Flop)

Analysis of Clocked Sequential Circuits (with D Flip Flop)



Example 1: Step 2: State Table

$$D = Q_{n+1}$$

$$Q_A^+ = DA$$

$$Q_B^+ = DB$$

Analysis of Clocked Sequential Circuits (with D Flip Flop)

P.S.			N.S.		
Q_A	Q_B	x	Q_A^+	Q_B^+	y
0	0	0	0	0	1
0	0	1	0	0	0
0	1	0	1	1	0
0	1	1	1	1	0
1	0	0	0	0	1
1	0	1	1	0	1
1	1	0	1	0	0
1	1	1	1	0	1

$$Q_B^+ = DB = \bar{A}AQ_B$$

$$= 1 \cdot 0 = 0$$

$$y = 1 \cdot 1 + 0 \cdot 0 = 1$$

$$y = 0 \cdot 1 + 1 \cdot 0 = 0$$

Analysis of Clocked Sequential Circuits (with D Flip Flop)

Q_A	Q_B	x	Q_A	Q_B	y
0	0	0	0	0	1
0	0	1	0	0	0
0	1	0	1	1	0
0	1	1	1	1	0
1	0	0	0	0	1
1	0	1	1	0	1
1	1	0	1	0	0
1	1	1	1	0	1

$$\begin{aligned}
 S_0 &= \begin{matrix} Q_A & Q_B \\ 0 & 0 \end{matrix} & S_2 &= 10 \\
 S_1 &= \begin{matrix} Q_A & Q_B \\ 0 & 1 \end{matrix} & S_3 &= 11
 \end{aligned}$$

$$= 1 \cdot 0 = 0$$

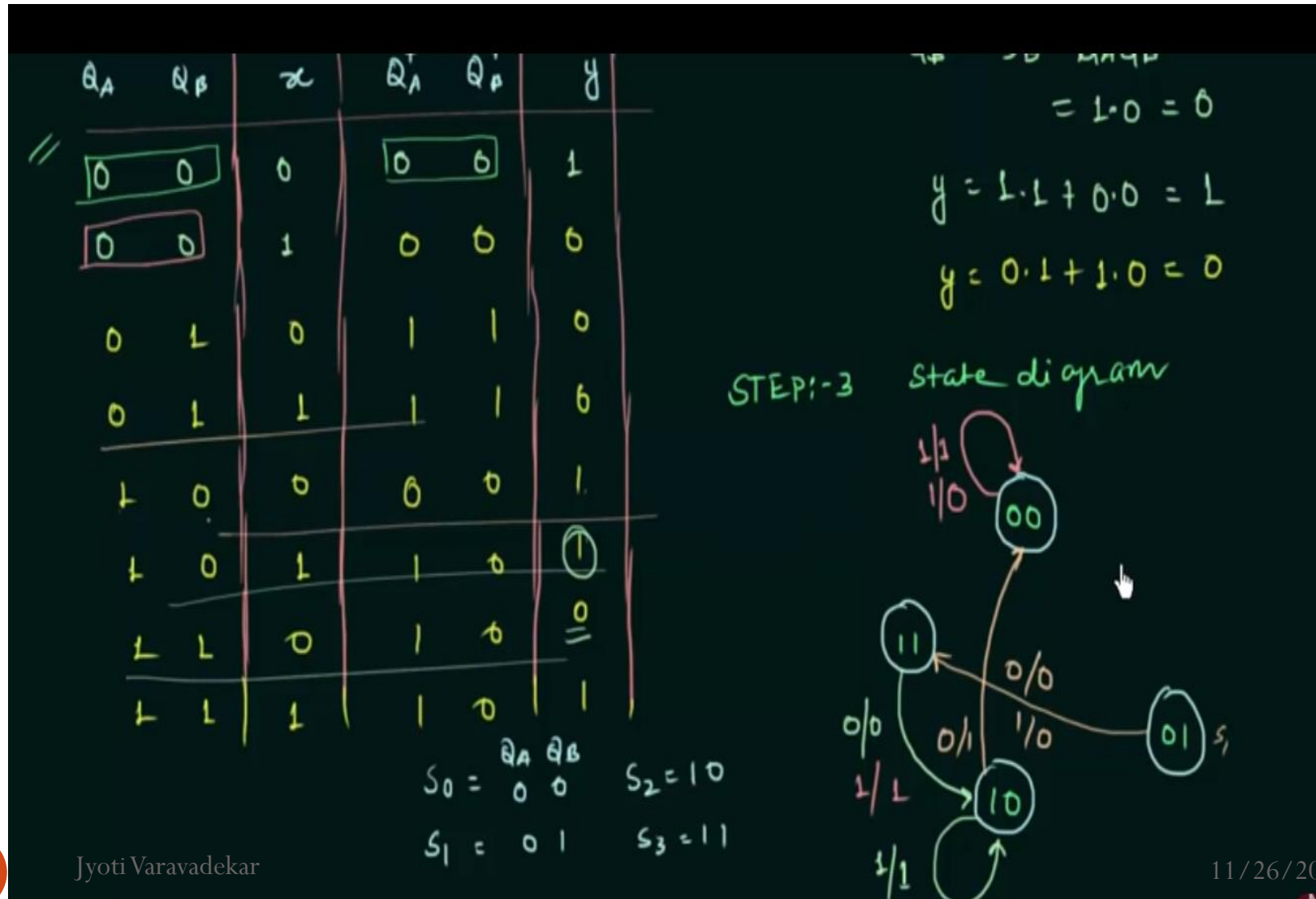
$$y = 1 \cdot 1 + 0 \cdot 0 = 1$$

$$y = 0 \cdot 1 + 1 \cdot 0 = 0$$

STEP:-3 state diagram

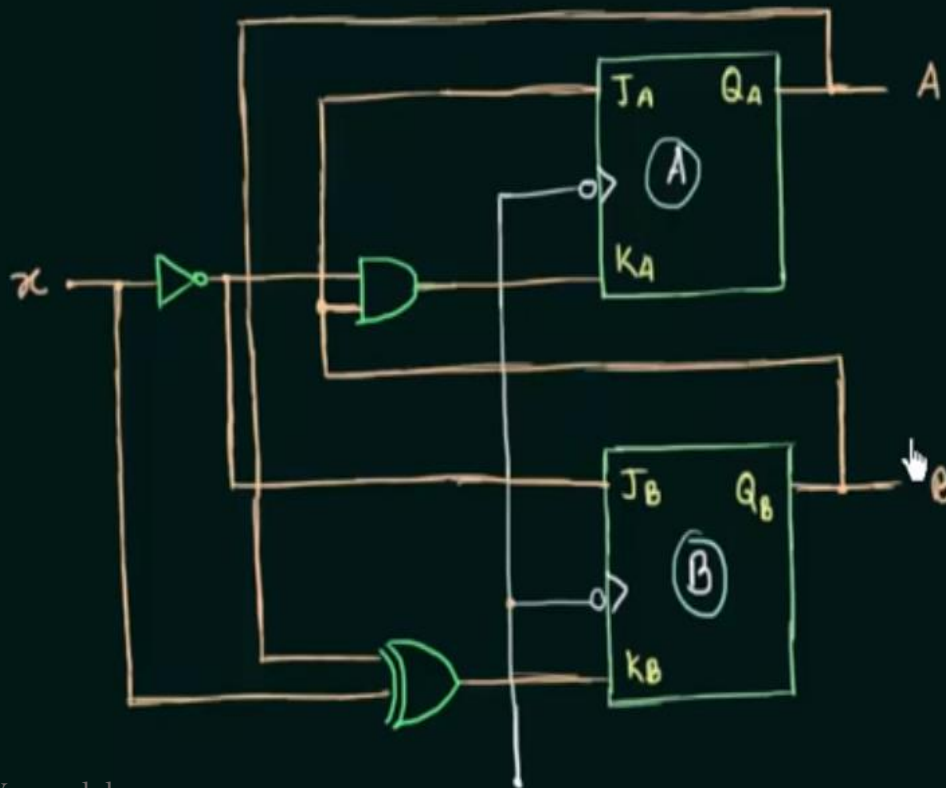


Example 1: Step 3: State Diagram



Example2

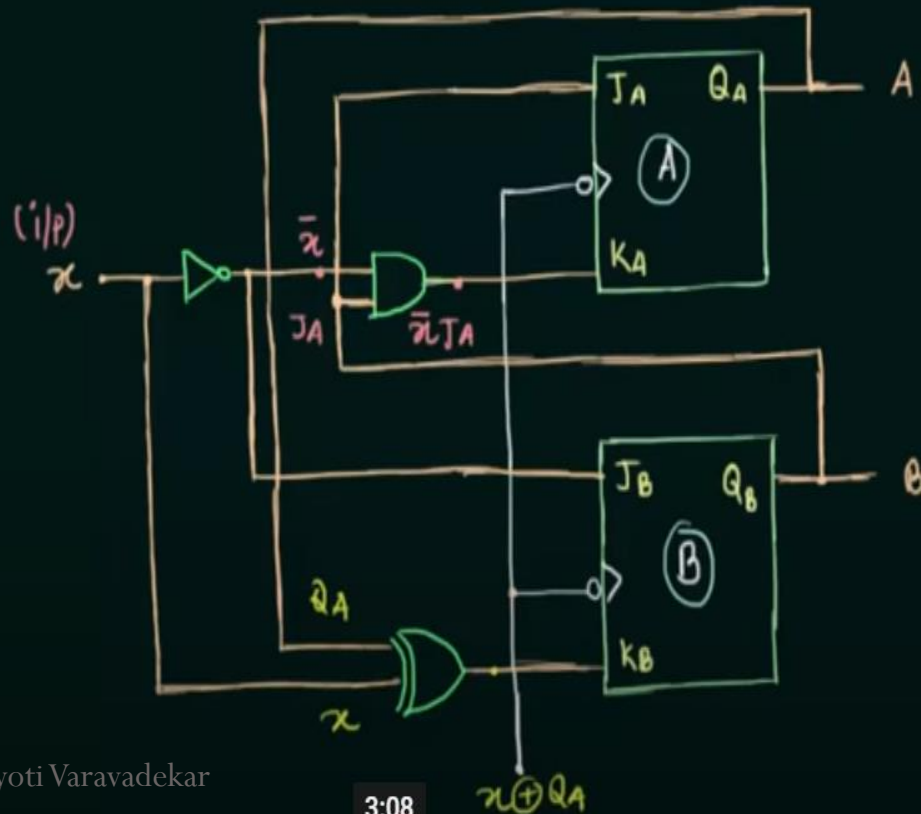
Analysis of Clocked Sequential Circuits (with Jk Flip Flop)



Example2 : Step 1: I/P equations

Analysis of Clocked Sequential Circuits (with JK Flip Flop)

Analysis of Clocked Sequential Circuits (with Jk Flip Flop)



STEP:-1 i/p cq^n

$$\uparrow J_A = Q_B \quad K_A = \bar{\alpha} J_A$$

$$J_B = \bar{x} \quad K_B = x \oplus Q_A$$

$$= \bar{x}Q_A + \bar{Q}_Ax$$

Example 2 : Step 2: State Table

State table :-

Present State

Q_A	Q_B	x	J_A	K_A	J_B	K_B	Q_A^+	Q_B^+
0	0	0						
0	0	1						
0	1	0						
0	1	1						
1	0	0						
1	0	1						
1	1	0						
1	1	1						

$$J_A = Q_B \quad K_A = \bar{x} J_A$$

$$J_B = \bar{x} \quad K_B = x \oplus Q_A$$

$$= \bar{x} Q_A + \bar{Q}_A x$$

Example2 : Step 2:State Table

Analysis of Clocked Sequential Circuits (with JK Flip Flop)

Step:- 2

State table :-

$$J_A = Q_B$$

$$J_B = \bar{x}$$

$$K_A = \bar{x}J_A$$

$$K_B = x \oplus Q_A$$

$$= \bar{x}Q_A + \bar{Q}_A x$$

→ odd 1's detect

Present State		i/p					N.S.	
Q_A	Q_B	x	J_A	K_A	J_B	K_B	Q_A^+	Q_B^+
0	0	0	0	0	1	0	0	1
0	0	1	0	0	0	1	0	0
0	1	0	1	1	1	0	1	1
0	1	1	1	0	0	1	1	0
1	0	0	0	0	1	1	1	1
1	0	1	0	0	0	0	1	0
1	1	0	1	1	1	1	0	0
1	1	1	1	0	0	0	1	1

Analysis of Clocked Sequential Circuits (with JK Flip Flop)

Present State i/p							N.S.		
Q_A	Q_B	κ	J_A	K_A	J_B	K_B	Q_A^+	Q_B^+	
0	0	0	0	0	1	0	0	1	$S_0 = 00$
0	0	1	0	0	0	1	0	0	$S_1 = 01$
0	1	0	1	1	1	0	1	1	$S_2 = 10$
0	1	1	1	0	0	1	1	0	$S_3 = 11$
1	0	0	0	0	1	1	1	1	
1	0	1	0	0	0	0	1	0	
1	1	0	1	1	1	1	0	0	
1	1	1	1	0	0	0	1	1	

Example 2 : Step 3: State Diagram

Analysis of Clocked Sequential Circuits (with JK Flip Flop)

Present State i/p $Q_{n+1} = D$ N.S. $= \bar{x}Q_A + \bar{Q}_Ax$

Q_A	Q_B	x	J_A	K_A	J_B	K_B	Q_A^+	Q_B^+
0	0	0	0	0	1	0	0	1
0	0	1	0	0	0	1	0	0
0	1	0	1	1	1	0	1	1
0	1	1	1	0	0	1	0	0
1	0	0	0	0	1	1	1	1
1	0	1	0	0	0	0	1	0
1	1	0	1	1	1	1	0	0
1	1	1	1	0	0	0	1	1

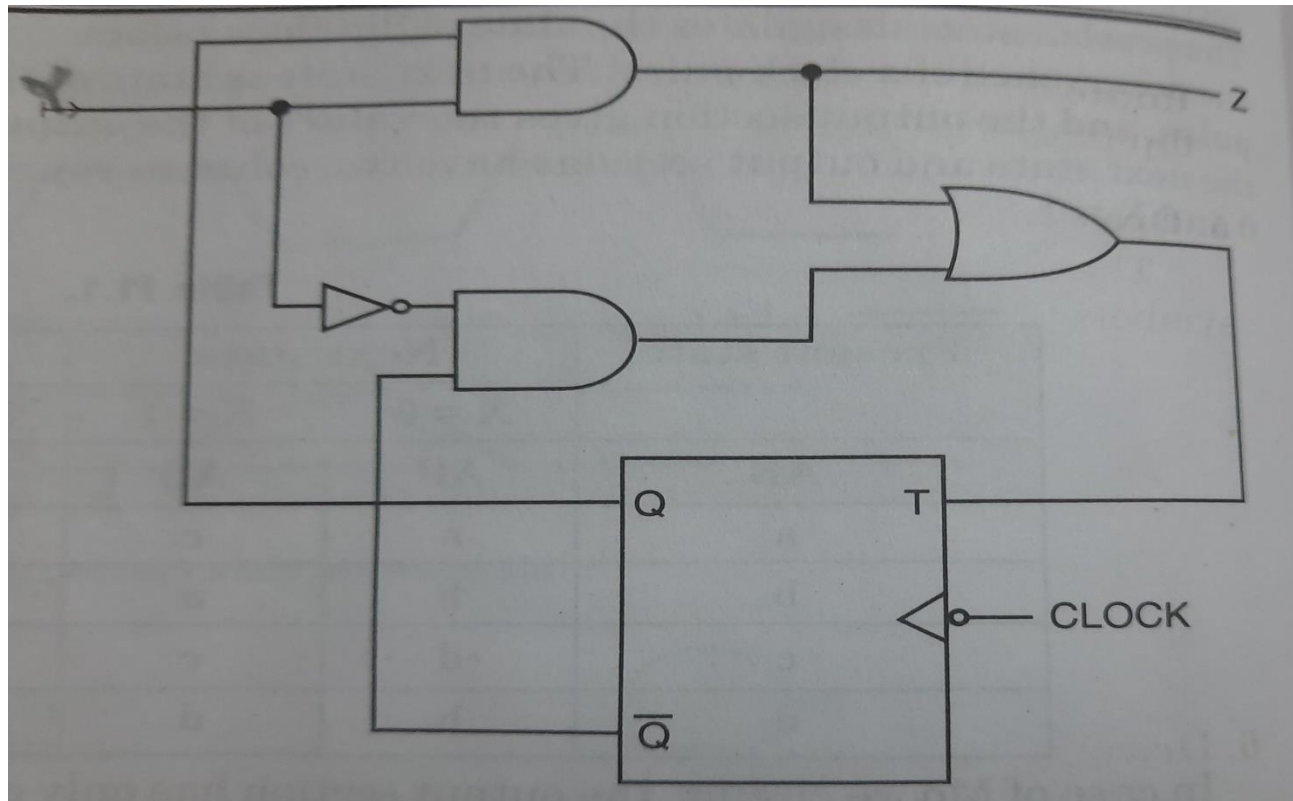
$Q_A Q_B$
 $S_0 = 00$
 $S_1 = 01$
 $S_2 = 10$
 $S_3 = 11$

```

graph TD
    00((00)) -- 0 --> 00
    00 -- 1 --> 01((01))
    01 -- 0 --> 00
    01 -- 1 --> 10((10))
    10 -- 0 --> 11((11))
    10 -- 1 --> 10
    11 -- 0 --> 00
    11 -- 1 --> 11
    
```

Analysis of clocked Sequential circuits

- Example 3



- Step1: Determine FF input equations and the output equations from the given sequential circuit

$$\begin{aligned} Z &= \alpha Q \\ T &= \alpha Q + \alpha' Q' \end{aligned}$$

- Step 2: Derive the transition equation:

$$\begin{aligned} Q^+ &= T \oplus Q \\ Q^+ &= (\alpha Q + \alpha' Q') \oplus Q \end{aligned}$$

3. Plot the next step map for each flip-flop

For Q^+

	0	1
0	1	0
1	1	0

$$Q^+ = (\rightarrow Q + \rightarrow Q) \quad Q$$

Fig. 11.10.

4. Plot the transition table

Present state	Next state		Output	
	$\alpha = 0$	$\alpha = 1$	$\alpha = 0$	$\alpha = 1$
Q	Q^+	Q^+	Z	Z
0	1	0	0	0
1	1	0	0	1

Important Point: If circuit consists of more than one flip-flop, we have to combine the flip-flop to derive the transition table (Refer next example).

5. Draw the state table

The above transition table can be converted into state table as shown in table 11.4. symbols to binary codes are assigned. They are $a = 0$, $b = 1$.

Table 11.4. State table

Present state	Next state		Output	
	$\alpha = 0$	$\alpha = 1$	$\alpha = 0$	$\alpha = 1$
a(0)	b	a	0	0
b(1)	b	a	0	1

6. Draw state diagram

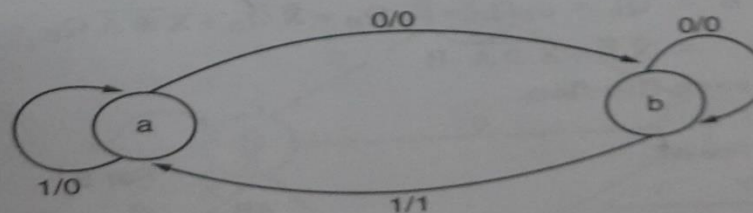
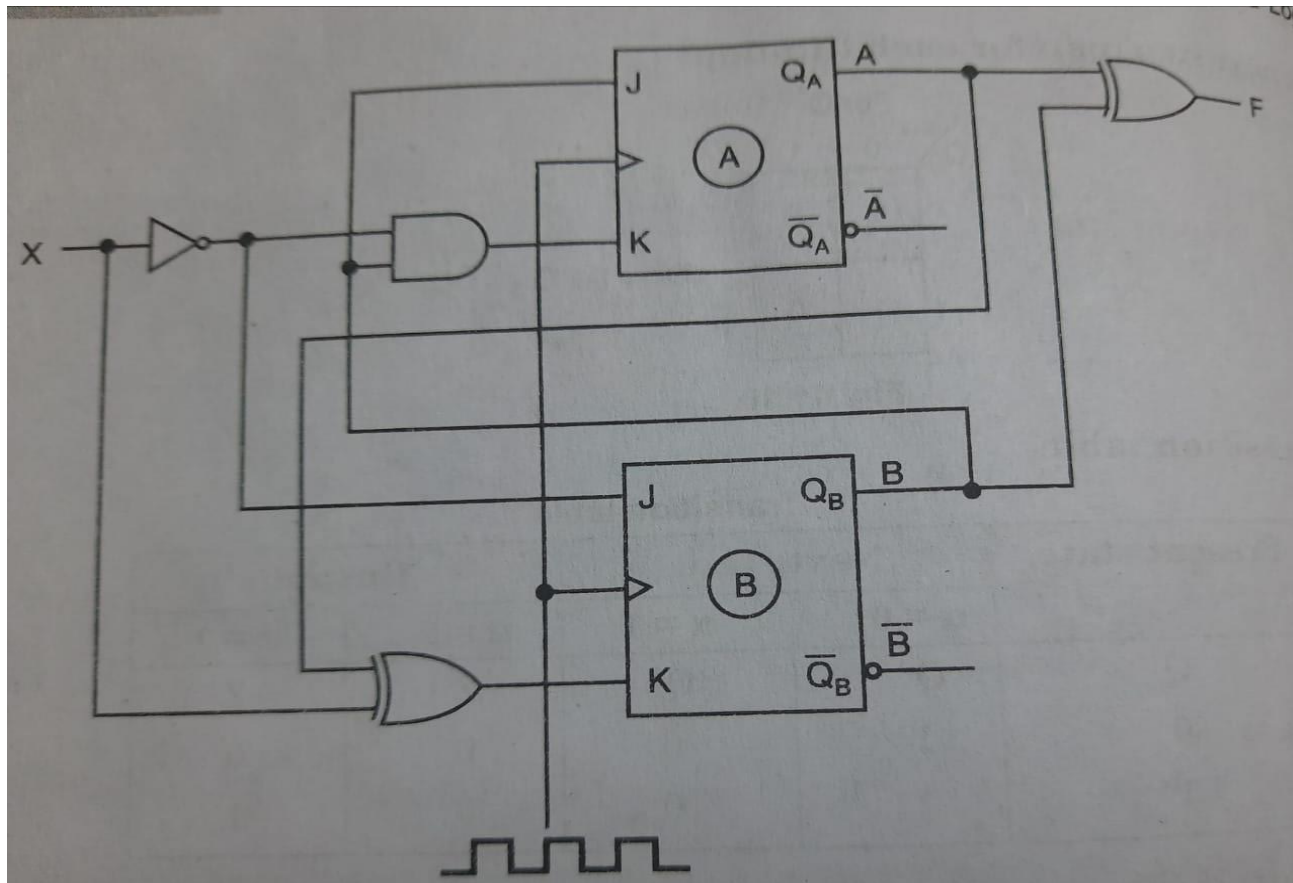


Fig. 11.11. State diagram

Example 4: Analyze the given circuit and construct transition table, state table and state diagram for the Moore sequential circuit



Solution: 1. Determine the flip-flop input equations and the output equations from the sequential circuit

$$F = A \oplus B$$

$$J_A = B, \quad K_A = \bar{X}B$$

$$J_B = \bar{X}, \quad K_B = X \oplus A$$

2. Derive the transition equations

The transition equations for JK flip-flops can be derived from the characteristic equation of JK flip-flop follows:

We know that for JK flip-flop

$$Q^+ = J\bar{Q} + \bar{K}Q$$

Thus,

$$\begin{aligned} A^+ = Q_A^+ &= J_A \bar{Q}_A + \bar{K}_A Q_A = B \bar{Q}_A + \bar{X}B Q_A = B \bar{Q}_A + (X + \bar{B})Q_A \\ &= B\bar{A} + (X + \bar{B})A \end{aligned}$$

and

$$\begin{aligned} B^+ = Q_B^+ &= J_B \bar{Q}_B + \bar{K}_B Q_B = \bar{X} \bar{Q}_B + \overline{X \oplus A} Q_B \\ &= \bar{X} \bar{B} + \overline{X \oplus A} \cdot B \end{aligned}$$

3. Plot a next-state maps for each flip-flop.

The next-state maps are

For A^+

		X	
	AB	0	1
00		0	0
01		1	1
10		1	1
11		0	1

$A^+ = B\bar{A} + (X + \bar{B})A$

Binary sequence ↗

For B^+

		X	
	AB	0	1
00		1	0
01		1	0
10		1	0
11		0	1

Binary sequence ↗

$$B^+ = \bar{X} \bar{B} + \overline{X \oplus A} \cdot B$$

Fig. 11.13.

4. Plot the transition table

The transition table can be formed by combining the above two maps. The table 11.5 shows the transition table.a

and maps. The table 11.5 shows the

Present state		Next state				Output
		X = 0		X = 1		
A	B	A ⁺	B ⁺	A ⁺	B ⁺	F = (A ⊕ B)
0	0	0	1	0	0	0
0	1	1	1	1	0	1
1	0	1	1	1	0	1
1	1	0	0	1	1	0

at point: For Moore sequential

Important point: For Moore sequential circuit, output only depends on present state and not on the input.

5. Draw the state table

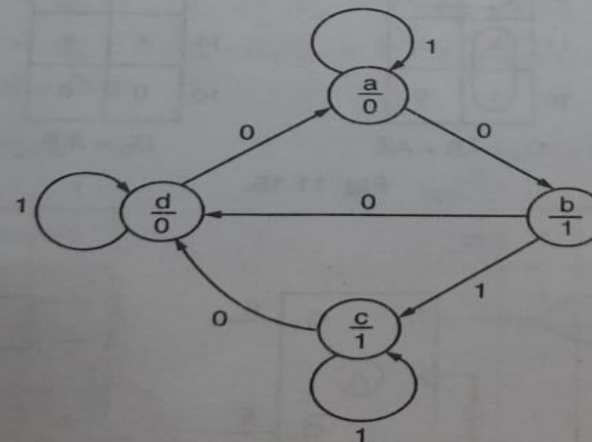
By assigning a = 00, b = 01, c = 10 and d = 11, we can write state table from the transition table as shown below.

Table 11.6.

Present state A B	Next state		output F
	X = 0	X = 1	
	A ⁺ B ⁺	A ⁺ B ⁺	
a	b	a	0
b	d	c	1
c	d	c	1
d	a	d	0

6. Draw state diagram

From the state table, we can draw state diagram as shown in figure 11.14.



State reduction technique

- Avoids redundant states
- **Redundant states** - If every possible set of inputs generate exactly same output and same next state
- When two states are equivalent, one of them can be removed without changing input-output relationship

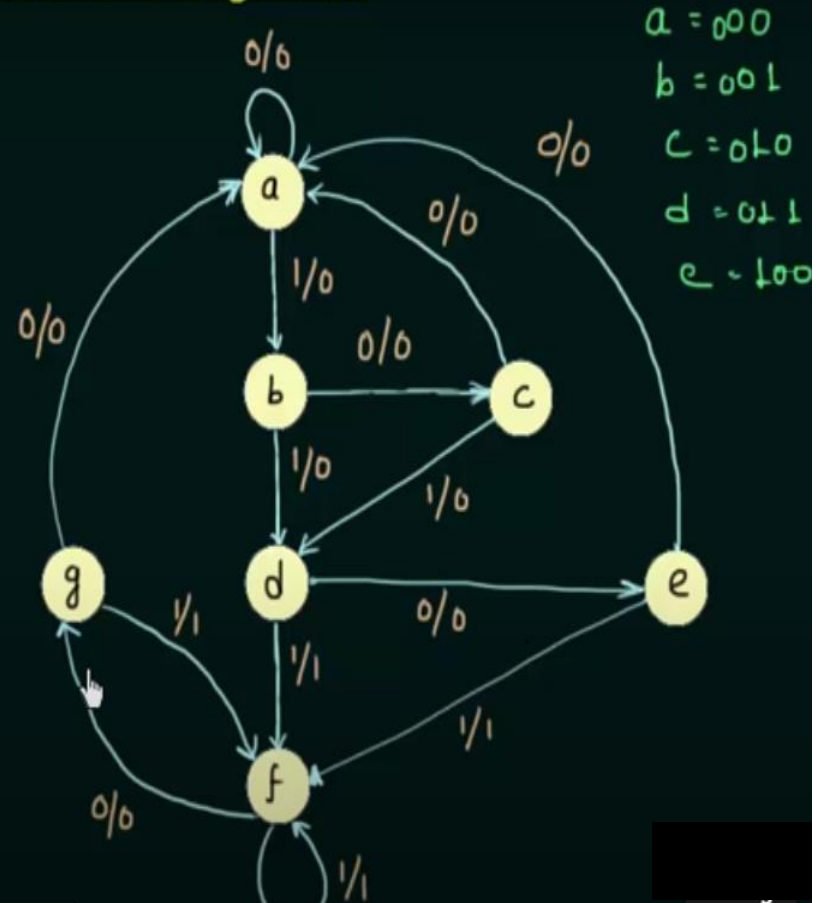
Advantages applying state reduction technique:

- Reduction in redundant states reduce requirement of FFs and logic gates
- Reduction in cost of final circuit
- Area reduction
- Power dissipation minimization

State Reduction and Assignment

State Reduction and Assignment

P.S.	N.S.		output	
	$x=0$	$x=1$	$x=0$	$x=1$
a	a	b	0	0
b	c	d	0	0
c	a	d	0	0
d	e	f	0	1
e	a	f	0	1
f	g	f	0	1
g	a	f	0	1



If next state and output of two present states are same then we can eliminate one state

State table: -

P.s.	N.s.		output	
	$x=0$	$x=1$	$x=0$	$x=1$
a	a	b	0	0
b	c	d	0	0
c	a	d	0	0
d	e	f	0	1
<u>e</u>	<u>a</u>	<u>f</u>	<u>0</u>	<u>1</u>
f	g	f	0	1
<u>g</u>	<u>a</u>	<u>f</u>	<u>0</u>	<u>1</u>

From state table it is found that

$$e=g$$

State table: -

P.s.	N.s.		output	
	$x=0$	$x=1$	$x=0$	$x=1$
a	a	b	0	0
b	c	d	0	0
c	a	d	0	0
d	e	f	0	1
<u>e</u>	<u>a</u>	<u>f</u>	<u>0</u>	<u>1</u>
f	g	f	0	1

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State table :-

P.S.	N.S.		output	
	$x=0$	$x=1$	$x=0$	$x=1$
a	a	b	0	0
b	c	d	0	0
c	a	d	0	0
Ⓓ	<u>e=g</u>	<u>f</u>	<u>0</u>	<u>1</u>
<u>e</u>	<u>a</u>	<u>f</u>	<u>0</u>	<u>1</u>
Ⓕ	<u>g</u>	<u>f</u>	<u>0</u>	<u>1</u>

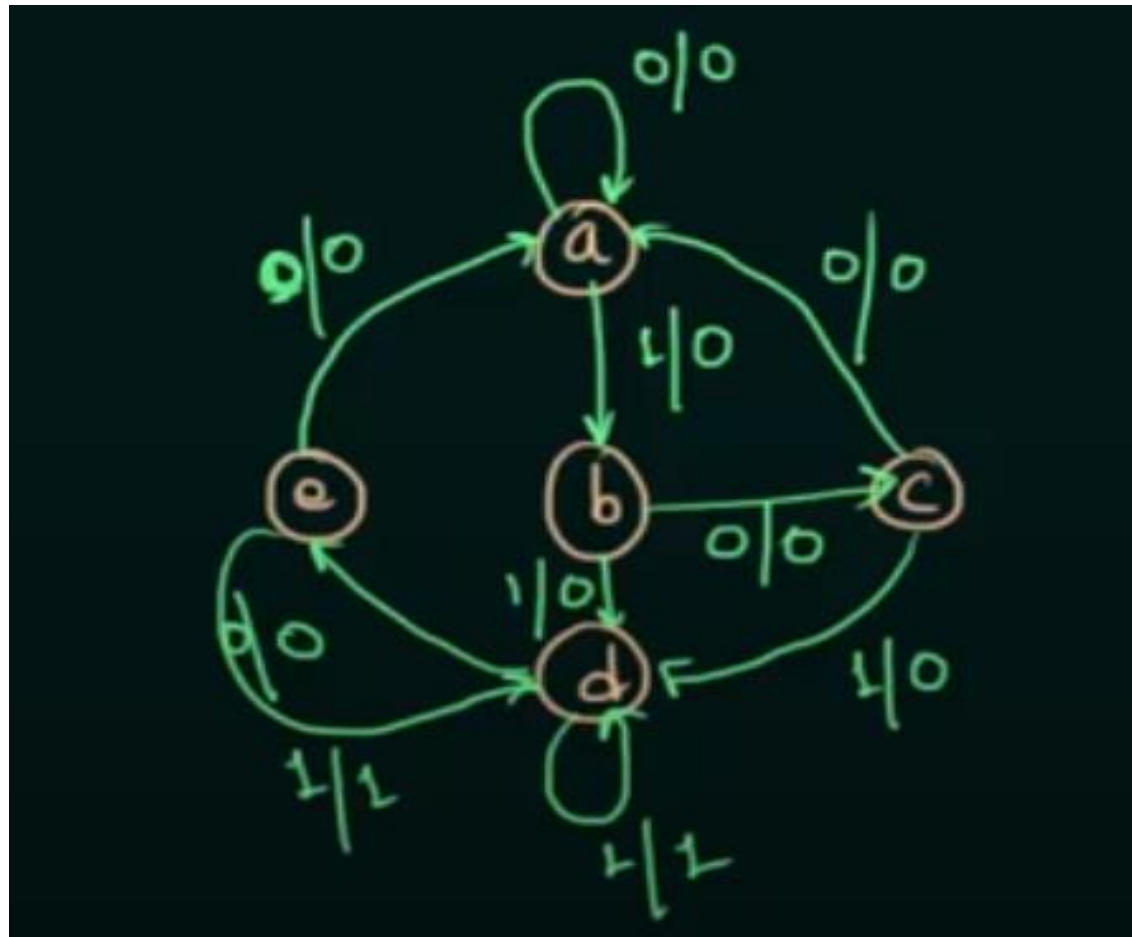
d=f

State table: -

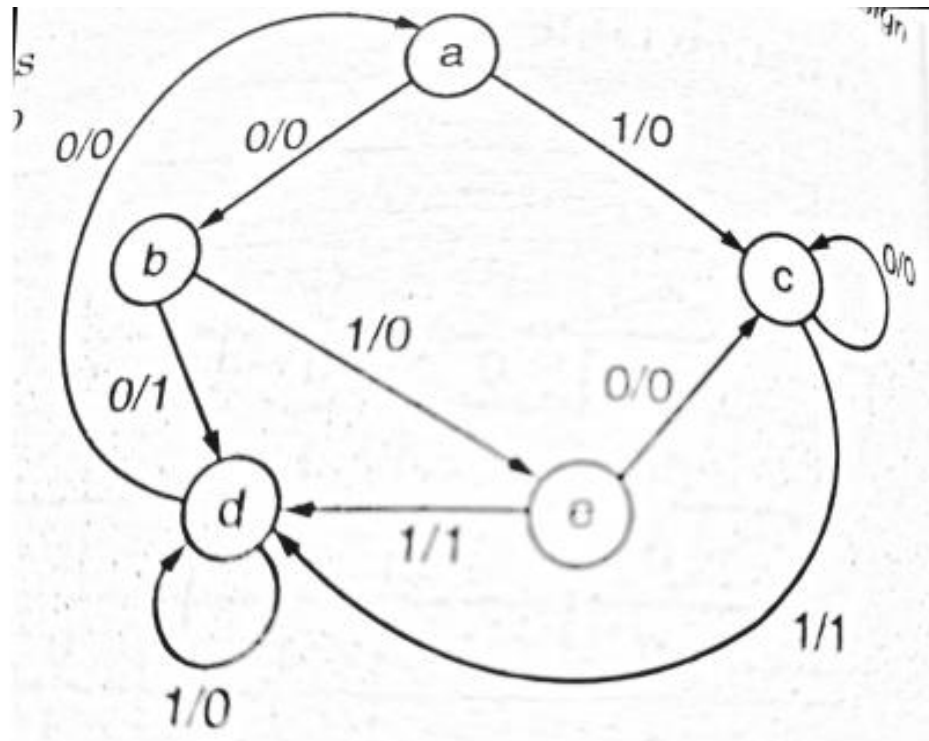
P.S.	N.S.		Output	
	$x=0$	$x=1$	$x=0$	$x=1$
a	a	b	0	0
b	c	d	0	0
c	a	d	0	0
Ⓓ	<u>e=g</u>	<u>f</u>	<u>0</u>	<u>1</u>
<u>e</u>	<u>a</u>	<u>f</u>	<u>0</u>	<u>1</u>

No further state reduction is possible

Reduced state diagram



State reduction example2



only input-output sequence. **Ans.** State table for given state diagram.

Step 1. Determine the state table for given state diagram.

Table 11.25 shows the state table for given state diagram

Table 11.25. State table

Present state	Next state		Output	
	X = 0	X = 1	X = 0	X = 1
a	b	c	0	0
b	d	e	1	0
c	[c	d	0	1]
d	a	d	0	0
e	[c	d	0	1]

Step 2 : Find equivalent states.

Table 11.26. Reduced state table

Present state	Next state		Output	
	X = 0	X = 1	X = 0	X = 1
a	b	c	0	0
b	d	c	1	0
c	c	d	0	1
d	a	d	0	0

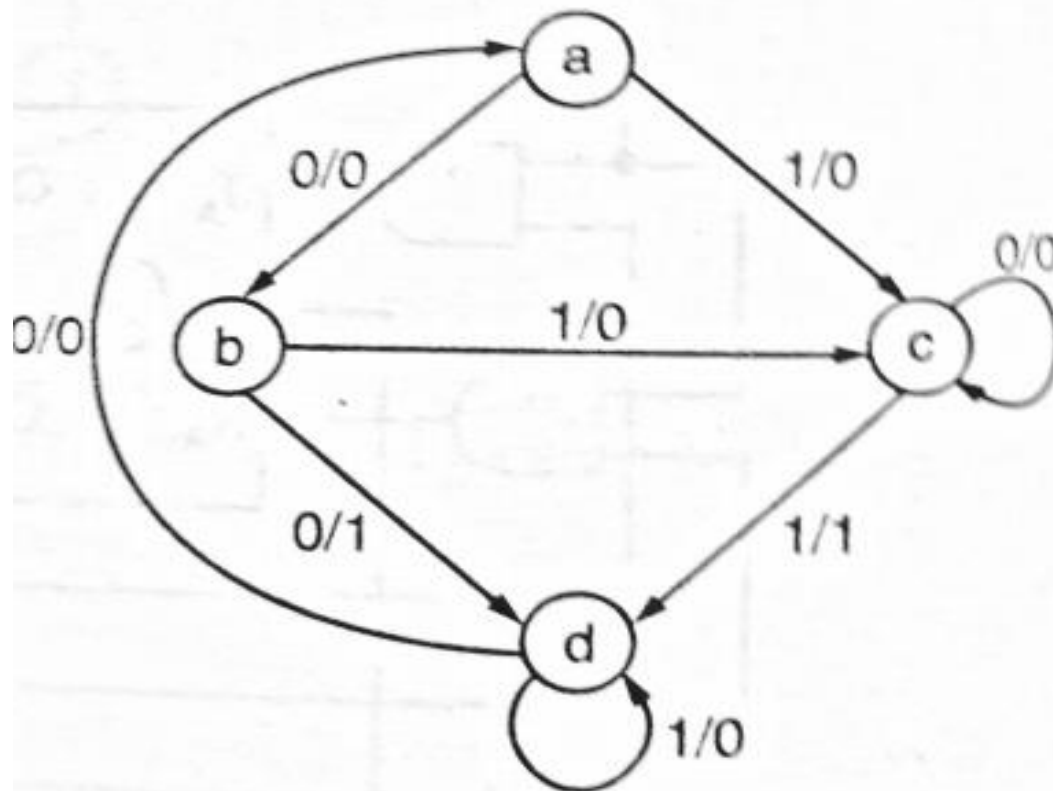


Fig. 11.55. Reduced state diagram

Design of clocked sequential circuits

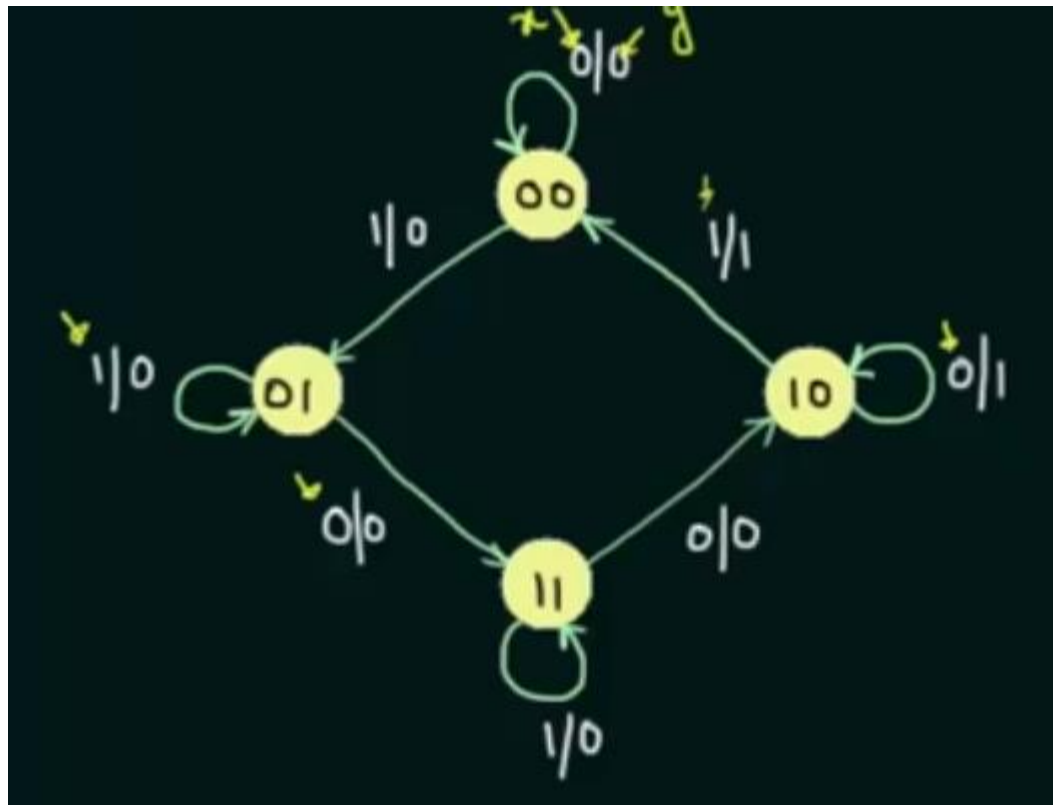
Design Procedure for Clocked Sequential Circuits

Design Procedure for Clocked Sequential Circuits

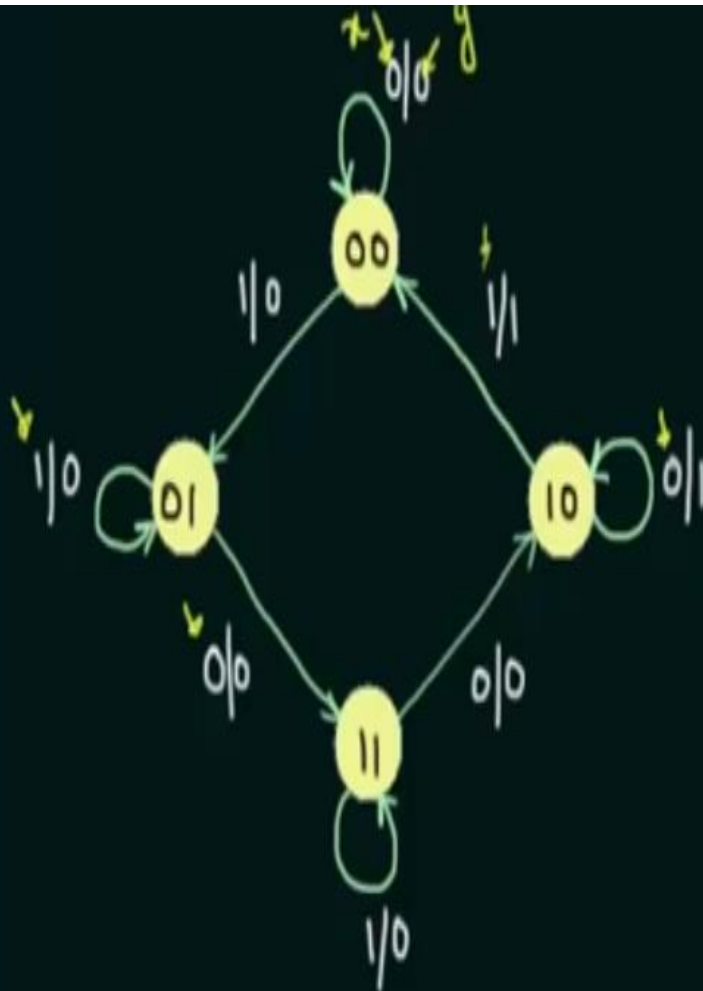
- Step 1: A State diagram or timing diagram is given, which describes the behaviour of the circuit that is to be designed.
- Step 2: Obtain the state table.
- Step 3: The number of states can be reduced by state reduction method.
- Step 4: Do state assignment. (If required)
- Step 5: Determine the number of flip-flops required and assign letter symbols.
- Step 6: Decide the type of flip-flop to be used.
- Step 7: Derive the circuit excitation table from state table.
- Step 8: Obtain the expression for circuit output and flip flop input.
- Step 9: Implement the circuit.

Design Example 1

- For the following given state diagram ,design a sequential circuit



State table preparation

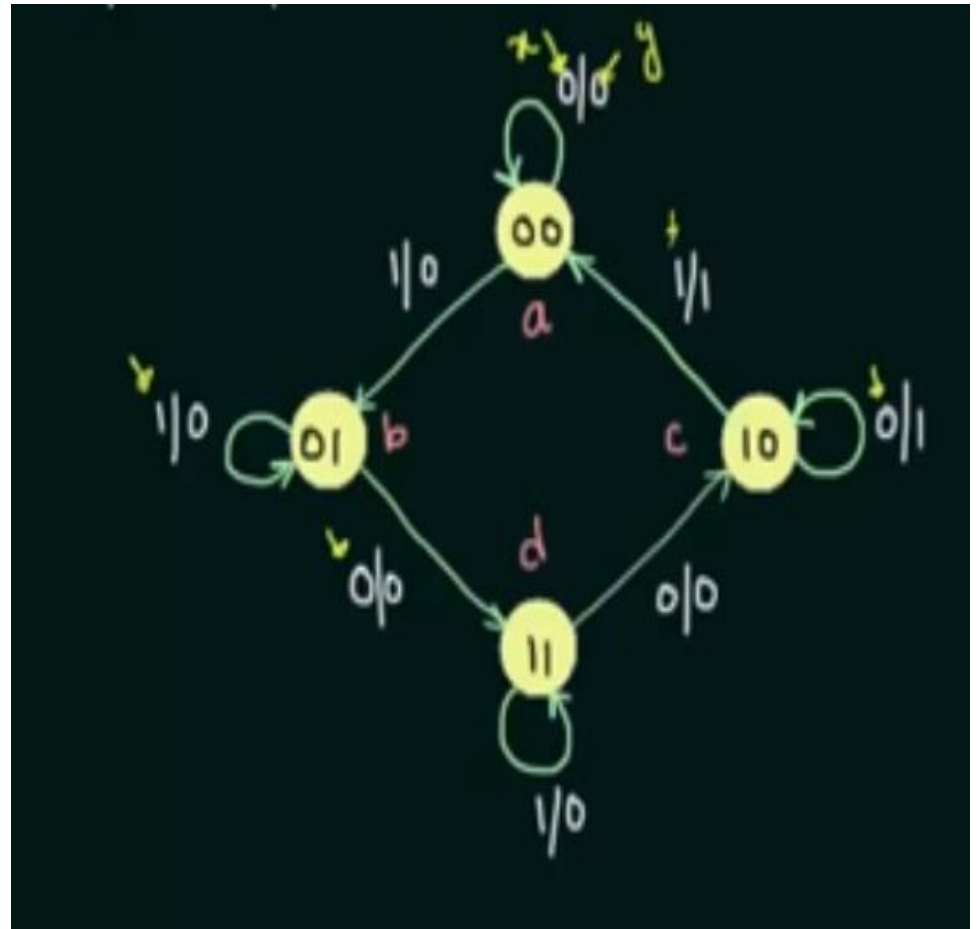


State table: -

P.S.		N.S.				Output (y)	
		x=0		x=1		x=0	x=1
Q _A	Q _B	Q _A ⁺	Q _B ⁺	Q _A ⁺	Q _B ⁺		
0	0	0	0	0	1	0	0
0	1	1	1	0	1	0	0
1	0	1	0	0	0	1	1
1	1	1	0	1	1	0	0

State assignment

$a = 00$
 $b = 01$
 $c = 10$
 $d = 11$



State excitation table

P.S.			N.S.		ff: p	y
Q_A	Q_B	x	Q_A^+	Q_B^+		
0	0	0	0	0		0
0	0	1	0	1		0
0	1	0	1	1		0
0	1	1	0	1		0
1	0	0	1	0		1
1	0	1	0	0		1
1	1	0	1	0		0
1	1	1	1	1		0

Design Procedure for Clocked Sequential Circuits

clk	T	Q_{n+1}
0	x	Q_n
1	0	Q_n
1	1	$\bar{Q}_n$

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

$$T = Q_n \oplus Q_{n+1}$$

P.S.			N.S.		ff's		y
$\bar{Q}_A$	$\bar{Q}_B$	x	$\bar{Q}_A^+$	$\bar{Q}_B^+$	T_A	T_B	
0	0	0	0	0	0	0	0
0	0	1	0	1	0	1	0
0	1	0	1	1	1	0	0
0	1	1	0	1	0	0	0
1	0	0	1	0	0	0	1
1	0	1	0	0	1	0	1
1	1	0	1	0	0	1	0
1	1	1	1	1	0	0	0

$$T_A = \bar{Q}_A Q_B \bar{x} + Q_A \bar{Q}_B x$$

$$T_B = \bar{Q}_A \bar{Q}_B x + Q_A Q_B \bar{x}$$

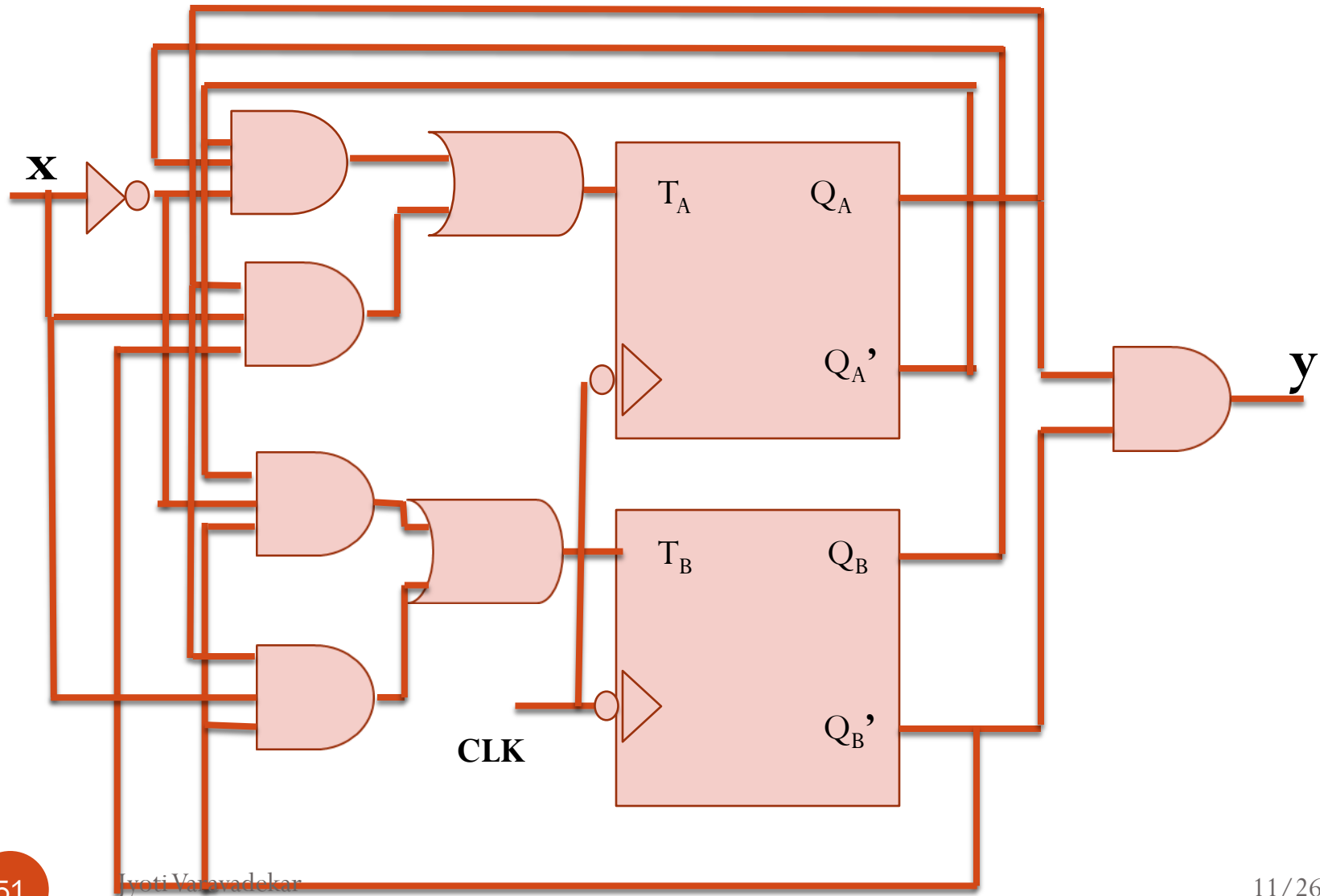
$$y = \underline{Q_A \bar{Q}_B \bar{x}} + \underline{Q_A \bar{Q}_B x}$$

$$= Q_A \bar{Q}_B (\bar{x} + x)$$

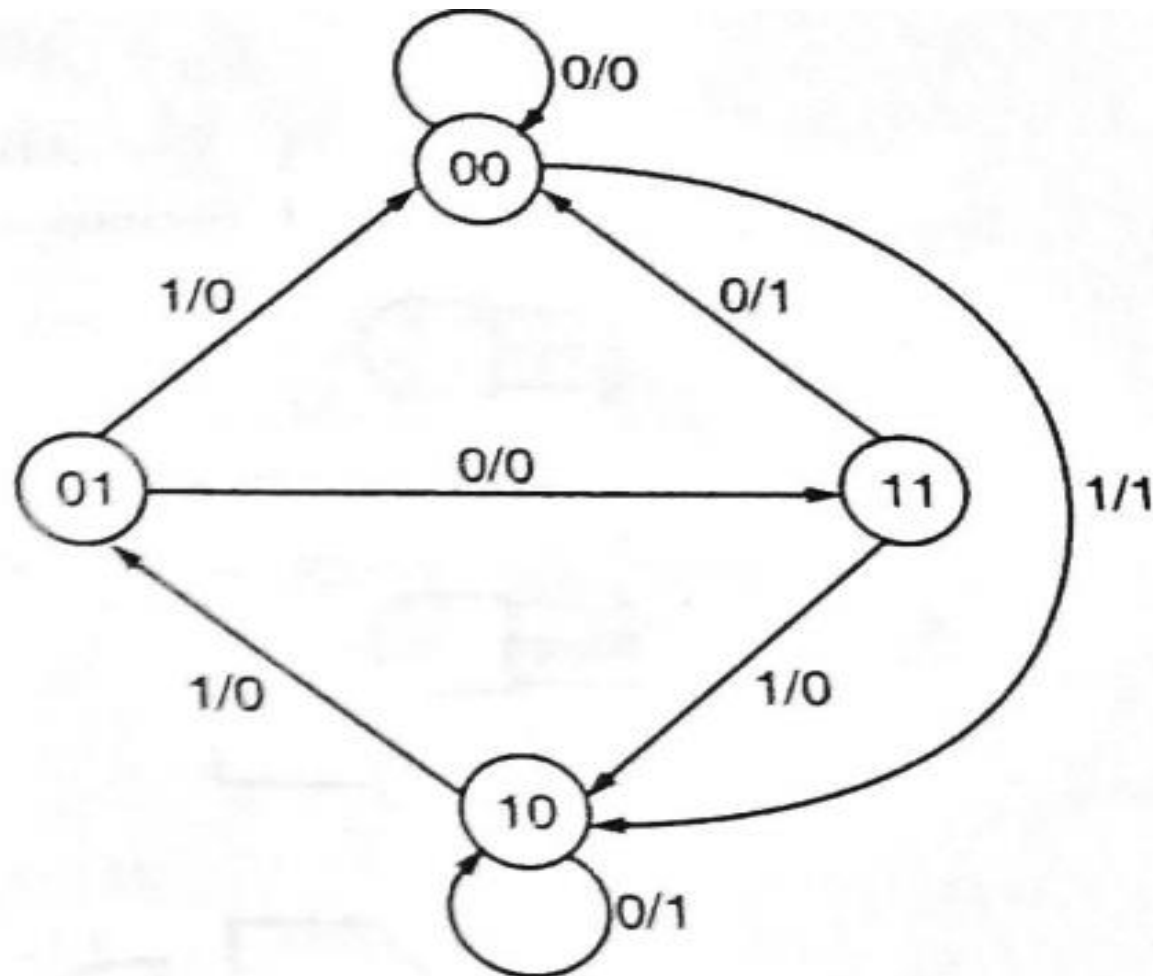
$$= Q_A \bar{Q}_B \cdot 1$$

$$y = Q_A \bar{Q}_B$$

Circuit Diagram



Question : Design synchronous sequential circuit for the given state diagram



Design example 2

for the next state and output functions. But first we have to decide on the type of flip-flops that will be used in the circuit. The most straightforward choice is to use D flip-flops, because in this case, the values of next state are simply clocked into the flip-flops to become the new values of present state. For other types of flip-flops, such as JK, T and RS, the relationship between the next-state variable and inputs to a flip-flop is not as straightforward as D flip-flop. For other types of flip-flops, we have to refer excitation table of flip-flop to find flip-flop inputs. This is illustrated in the following example.

EXAMPLE 11.5. A sequential circuit has one input and one output. The state diagram is shown in figure 11.24. Design the sequential circuit with

- D flip-flops
- T flip-flops
- RS flip-flops and
- JK flip-flops.

Solution: The state table for the state diagram shown in figure 11.24 is given in table 11.12.

Table 11.12.

Present state		Next state		Output	
		X = 0	X = 1	X = 0	X = 1
A	B	AB	AB	Y	Y
0	0	00	10	0	1
0	1	11	00	0	0
1	0	10	01	1	0
1	1	00	10	1	0

As seen from the state table, there are no equivalent states. Therefore, no reduction is the state diagram. The state table shows that circuit goes through four states, therefore, we require 2 flip-flops (number of states = 2^m , where m = number of flip-flops). Since two flip-flops are required first is denoted as A and second is denoted as B.

(i) Design using D flip-flops

As mentioned earlier, for D flip-flops next states are nothing but the new present states. Thus, we can directly use next states to determine the flip-flop input with the help of K-map simplification.

K-map simplification

X \ AB	00	01	11	10
0	0	1	0	1
1	1	0	1	0

(a) For flip-flop A

X \ AB	00	01	11	10
0	0	1	0	0
1	1	0	0	1

(b) For flip-flop B

X \ AB	00	01	11	10
0	0	0	0	1
1	1	0	0	0

(c) For output

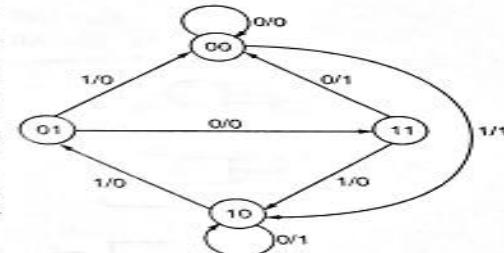


Fig. 11.24.

- Equations obtained after solving K map for D_A D_B and Y

$$D_A = A'B'X + A'BX' + ABX + AB'X'$$

$$D_B = A'BX' + AB'X$$

$$Y = A'B'X + AX'$$

$$D_A = \bar{A}BX + ABX$$

$$Y = \bar{A}BX + AX$$

With these flip-flop input functions and circuit output function, we can draw the logic diagram as follows.

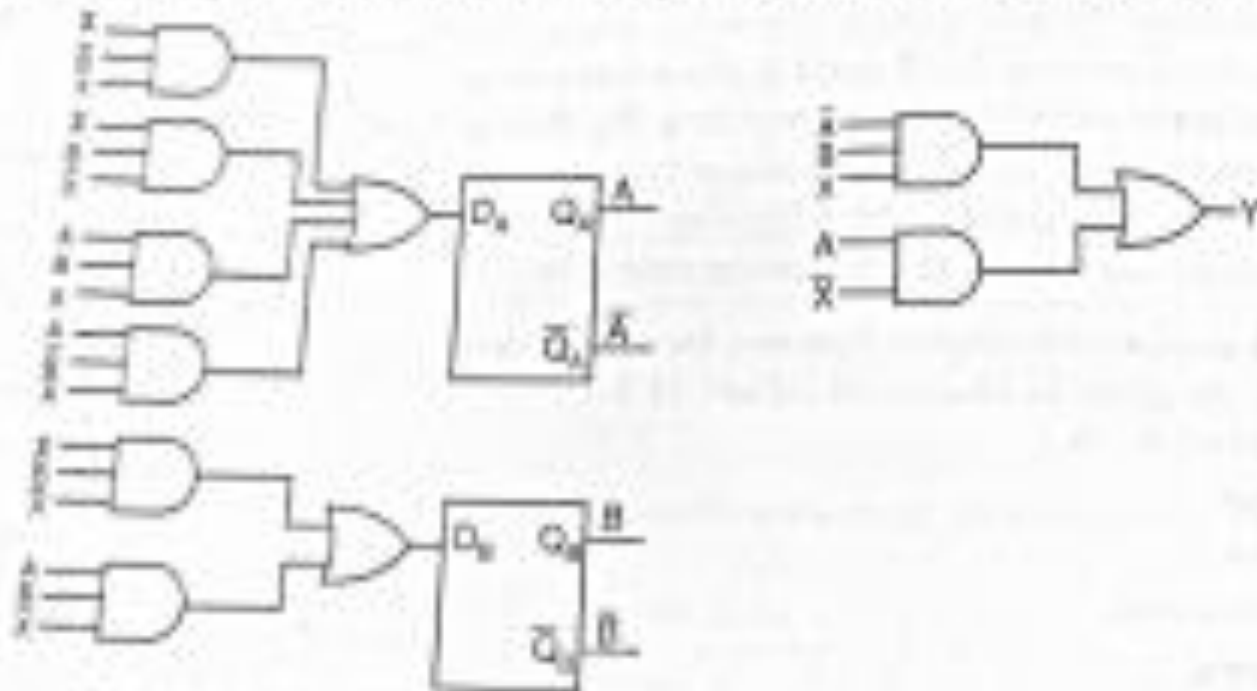


Fig. 11.26. Logic diagram of given sequential circuit using D flip-flop

(ii) Design using T flip-flops

Using the excitation table for T flip-flop shown in table 11.13, we can determine the excitation table for the given circuit as shown in table 11.14.

Table 11.13. Excitation table for T flip-flop

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

Table 11.14. Circuit excitation table

Present state		Input	Next state		Flip-flop inputs		Output
A	B	X	A	B	T_A	T_B	Y
0	0	0	0	0	0	0	0
0	0	1	1	0	1	0	1
0	1	0	1	1	1	0	0
0	1	1	0	0	0	1	0
1	0	0	1	0	0	0	1
1	0	1	0	1	1	1	0
1	1	0	0	0	1	1	1
1	1	1	1	0	0	1	0

The first row of circuit excitation table shows that there is no change in the state for both flip-flops. The transition from $0 \rightarrow 0$ for T flip-flop requires input T to be at logic 0. The second row shows that flip-flop A has transition $0 \rightarrow 1$. It requires the input T_A to be at logic 1. Similarly, we can find inputs for each flip-flop for each row in the table by referring present state, next state and excitation table.

Let us use K-map simplification to determine the flip-flop input functions and circuit output functions.

K-map simplification

AB \ X	0	1
00	0	1
01	1	0
11	1	0
10	0	1

(a) For flip-flop A

AB \ X	0	1
00	0	0
01	0	1
11	1	1
10	0	1

(b) For flip-flop B

AB \ X	0	1
00	0	1
01	0	0
11	1	0
10	1	0

(c) For output

Fig. 11.27.

Therefore, input function for

$$T_A = B\bar{X} + B\bar{X}$$

$$T_B = AB + BX + AX, \text{ and}$$

$$= A\bar{X} + ABX$$

Circuit output function

With these flip-flop input functions and circuit output function, we can draw the logic diagram as follows:

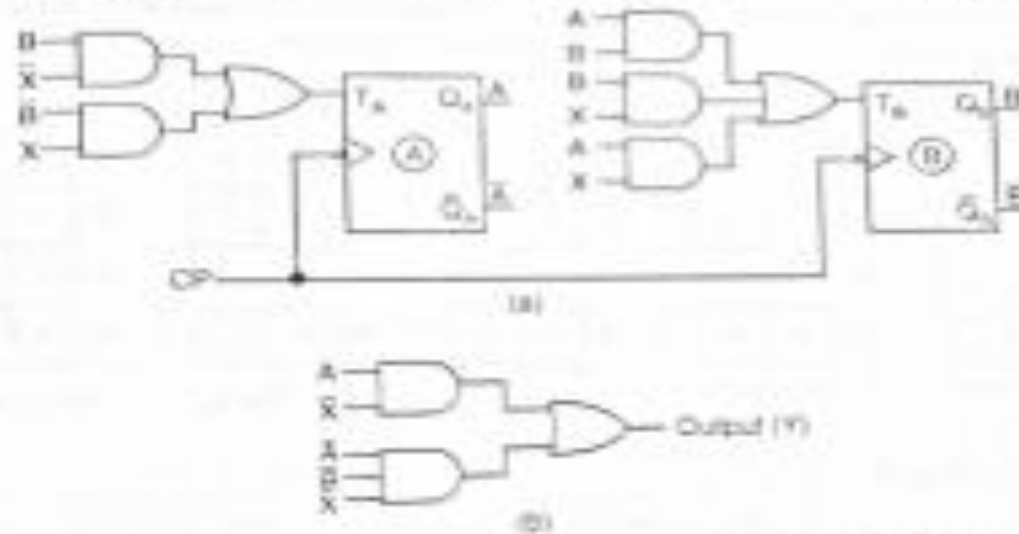
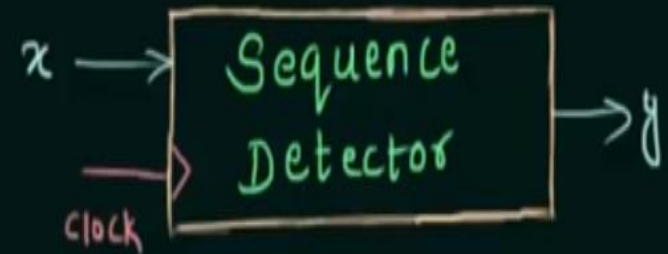


Fig. 11.28. Logic diagram of given sequential circuit using T flip-flops

Sequence or Pattern Detector

Pattern or Sequence Detector

- » The stream of bit has been feed as input, when the clock is high and a particular pattern/sequence is detected.
- » As soon as sequence is detected the output becomes high and then again becomes low.



$x = 0110010100\dots$

Sequence detector example 1

Sequence Detector (Example)

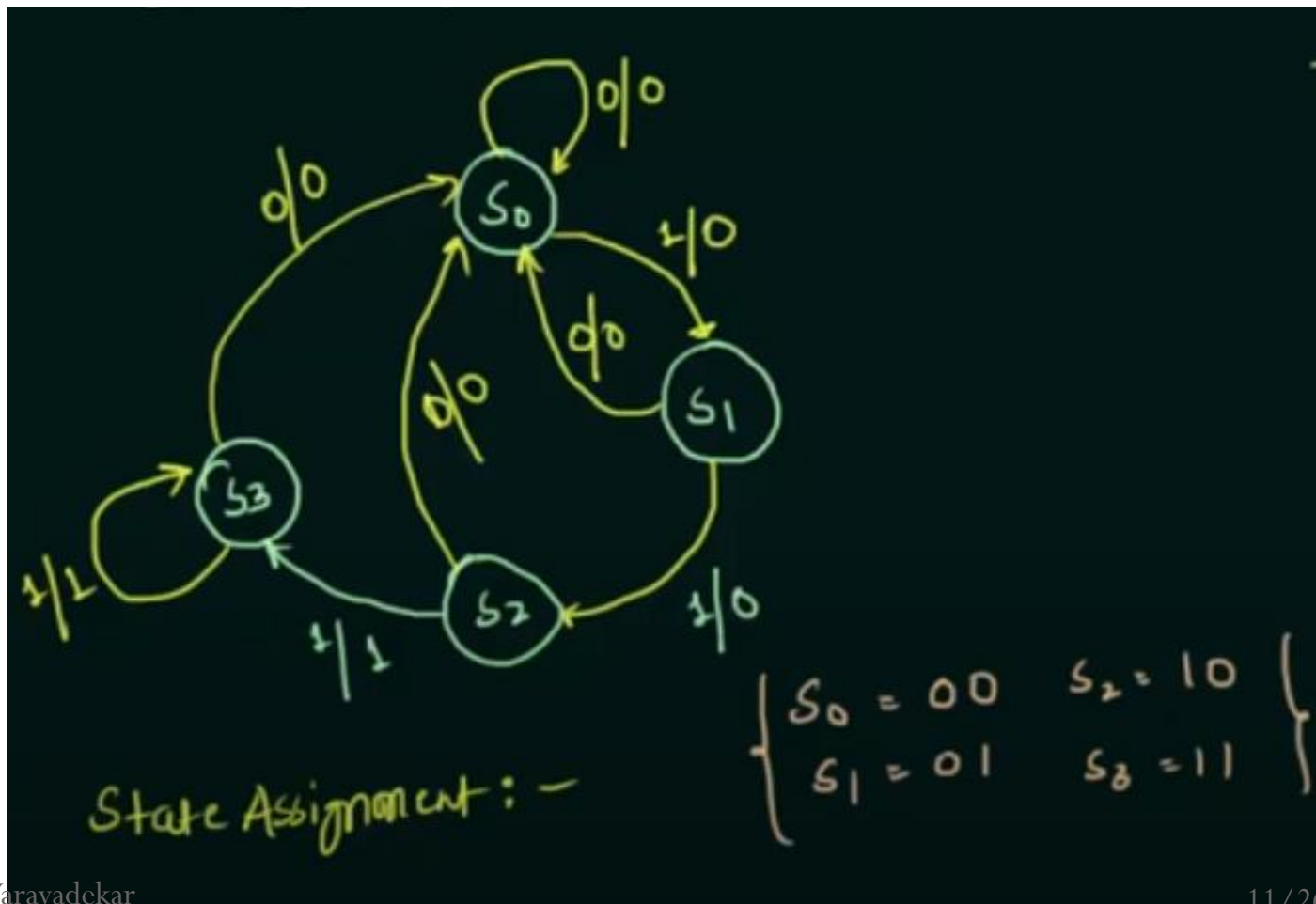
Pattern or Sequence Detector (Example)

Design a sequence detector to detect three or more consecutive 1's in a string of bits coming through an input line.

Example 1: Sequence detector

$x = 0011101110\dots$
 $y = 00001000110\dots$ (0V-)
 $y = 00001000100\dots$ (non ov.)

	Has	Awaits
$S_0 = \text{reset}$	---	LLL...
$S_1 = 1$	L	LL...
$S_2 = 11$	11	1-...



State Table

Next state equations for D FF

State Table :-

P.S.

Q_A	Q_B	x	Q_A^+	Q_B^+	y
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	0
0	1	1	1	0	0
1	0	0	0	0	0
1	0	1	1	1	1
1	1	0	0	0	0
1	1	1	1	1	1

$$N.S. = D$$

$$\textcircled{D_A} = Q_A^+$$

$$\textcircled{D_B} = Q_B^+$$

K map preparations for FF inputs and ckt O/P

Handwritten K-map for Q_A^+ (DA):

$Q_A \backslash Q_B Z$	00	01	11	10
0	0	0	1	0
1	0	1	1	0

Groupings: A vertical group of 1s at $Q_B Z = 11$ and a horizontal group of 1s at $Q_A = 1$.

$$Q_A^+ = D_A = Q_B Z + Q_A Z$$

$$= (Q_A + Q_B) \cdot Z$$

Handwritten K-map for Q_B^+ (DB):

$Q_A \backslash Q_B Z$	00	01	11	10
0	0	1	0	0
1	0	1	1	0

Groupings: A vertical group of 1s at $Q_B Z = 01$ and a horizontal group of 1s at $Q_A = 1$.

$$Q_B^+ = D_B = (\bar{Q}_B + Q_A) \cdot Z$$

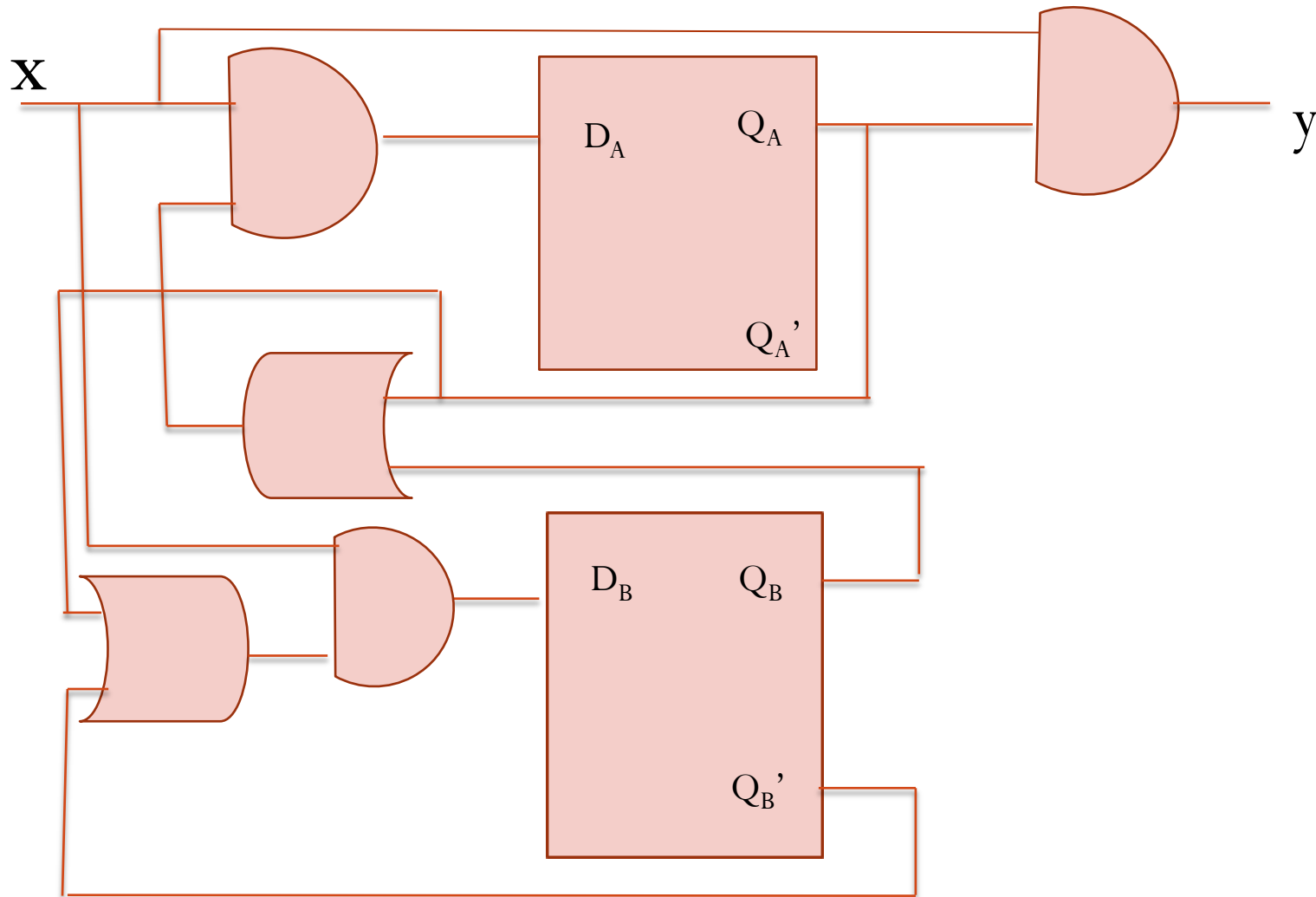
Handwritten K-map for output y :

$Q_A \backslash Q_B Z$	00	01	11	10
0	0	0	0	0
1	0	1	1	0

Groupings: A horizontal group of 1s at $Q_A = 1$.

∴ $y = Q_A Z$

Realization diagram

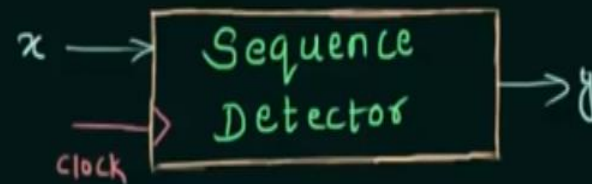


Example2:sequence detector

Sequence or Pattern Detector

Pattern or Sequence Detector

- >> The stream of bit has been feed as input, when the clock is high and a particular pattern/sequence is detected.
- >> As soon as sequence is detected the output becomes high and then again becomes low.



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$x = 0110010100\dots$

$y = 000001010$

010

- i) overlapping
- ii) do not cons. overlapping

Sequence or Pattern Detector

$x = 011001010 \dots$
 $y = 0000001010$

Annotations: s_1 is marked under the first '0' of the second sequence. A box highlights the sequence '010' in the first sequence, with arrows pointing to the corresponding '010' in the second sequence.

clock

010

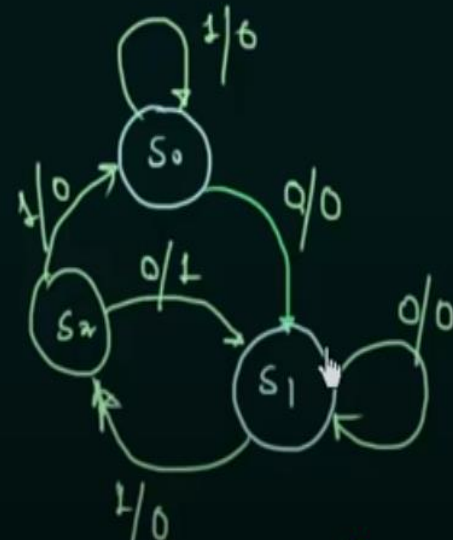
- overlapping
- do not cons. overlapping

Step-1 State diagram (Mealy machine)

$S_0 = \text{Reset (powerup)}$

$S_1 = 0$

$S_2 = 01$



$$\begin{array}{l} S_0 = \text{Reset (powerup)} \\ S_1 = 0 \\ S_2 = 01 \end{array}$$

Step-2 state assignment:-

$$S_0 = 00$$

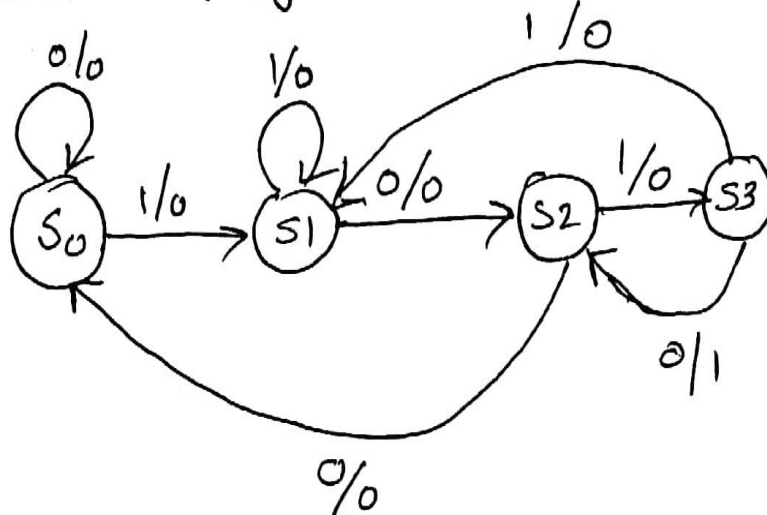
$$S_1 = 01$$

$$S_2 = 10$$

Example3 on sequence detector

- Design a sequence detector for the sequence 1010 using D flip flops and basic gates.
- The state diagram of a Mealy machine for a 1010 detector is:

State diagram for sequence 1010



Present state	Next state		output	
	X=0	X=1	X=0	X=1
S0	S0	S1	0	0
S1	S2	S1	0	0
S2	S0	S3	0	0
S3	S2	S1	1	0

- **State Assignment** Excitation Table with D flip flops as input F/Fs

- $Q_1Q_0=S_0=00$

- $Q_1Q_0=S_1=01$

- $Q_1Q_0=S_2=10$

- $Q_1Q_0=S_3=11$

Excitation Table

Present State		I/p	Next State		F/f I/ps		Output
Q ₁	Q ₀	X	Q ₁ *	Q ₀ *	D ₁	D ₂	Z
0	0	0	0	0	0	0	0
0	0	1	0	1	0	1	0
0	1	0	1	0	1	0	0
0	1	1	0	1	0	1	0
1	0	0	0	0	0	0	0
1	0	1	1	1	1	1	0
1	1	0	1	0	1	0	1
1	1	1	0	1	0	1	0

K-Maps

① To determine D1

$\Phi_0 X$	00	01	11	10
Φ_1 0	0	0	0	1
1	0	1	0	1

$$D1 = \Phi_0 \bar{X} + \Phi_1 \bar{\Phi}_0 X$$

② To determine D2

$\Phi_0 X$	00	01	11	10
Φ_1 0	0	1	1	0
1	0	1	1	0

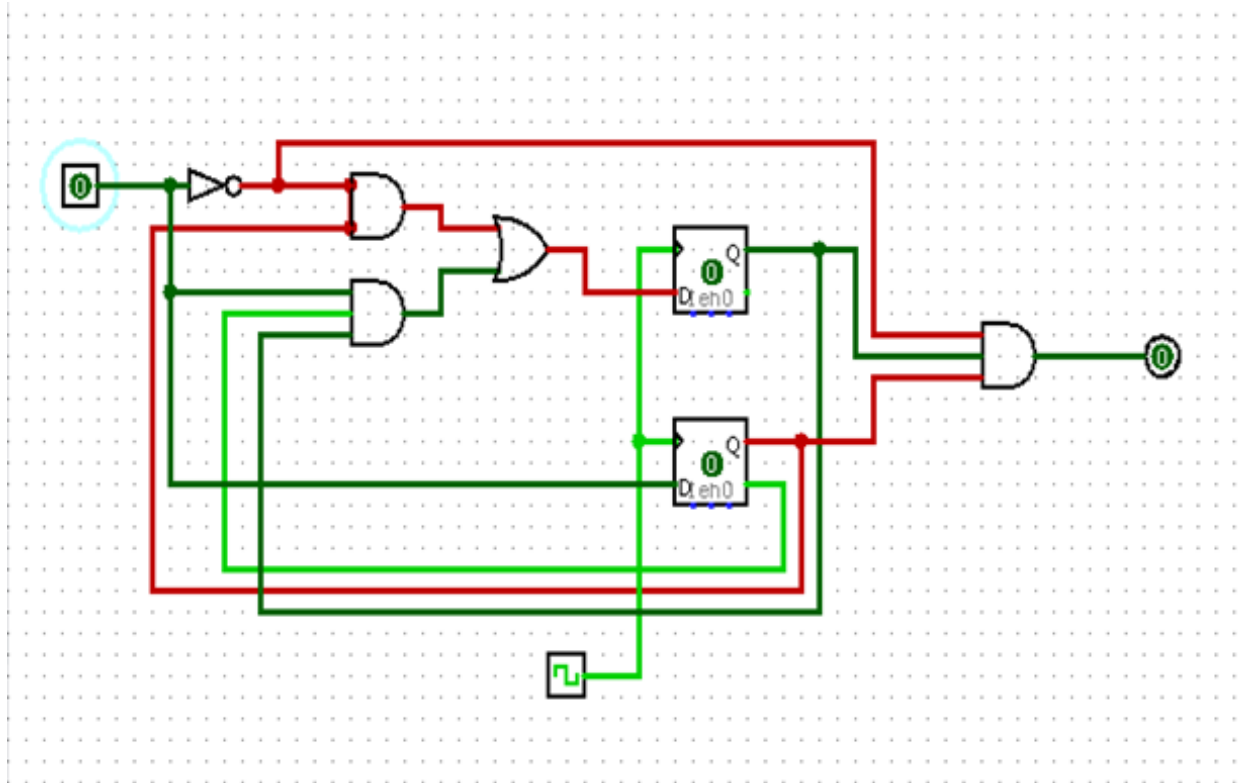
$$D2 = X$$

③ To determine Z

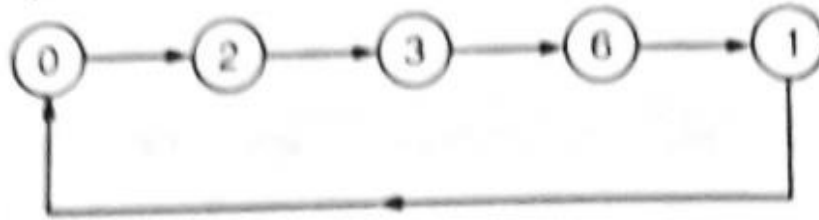
$$Z = \Phi_1 \Phi_0 \bar{X}$$

$\Phi_0 X$	00	01	11	10
Φ_1 0	0	0	0	0
1	0	0	0	1

Implementation in Logisim



Design a non-sequential counter using J-K flip-flop; as per following state diagram



Solution: 1. Excitation Table

Present State			Next State			Flip-flop Inputs					
Q_A	Q_B	Q_C	Q'_A	Q'_B	Q'_C	J_A	K_A	J_B	K_B	J_C	K_C
0	0	0	0	1	0	0	x	1	x	0	x
0	0	1	0	0	0	0	x	0	x	x	1
0	1	0	0	1	1	0	x	x	0	1	x
0	1	1	1	1	0	1	x	x	0	x	1
1	0	0	x	x	x	x	x	x	x	x	x
1	0	1	x	x	x	x	x	x	x	x	x
1	1	0	0	0	1	x	1	x	1	1	x
1	1	1	x	x	x	x	x	x	x	x	x

K-map Simplification

For J_A

Q_A	$Q_B Q_C$			
	00	01	11	10
0	0	0	1	0
1	X	X	X	X

$$J_A = Q_B Q_C$$

For K_A

Q_A	$Q_B Q_C$			
	00	01	11	10
0	X	X	X	X
1	X	X	X	1

$$K_A = 1$$

For J_B

Q_A	$Q_B Q_C$			
	00	01	11	10
0	1	0	X	X
1	X	X	X	X

$$J_B = \bar{Q}_C$$

For K_B

Q_A	$Q_B Q_C$			
	00	01	11	10
0	X	X	0	0
1	X	X	X	1

$$K_B = Q_A$$

For J_C

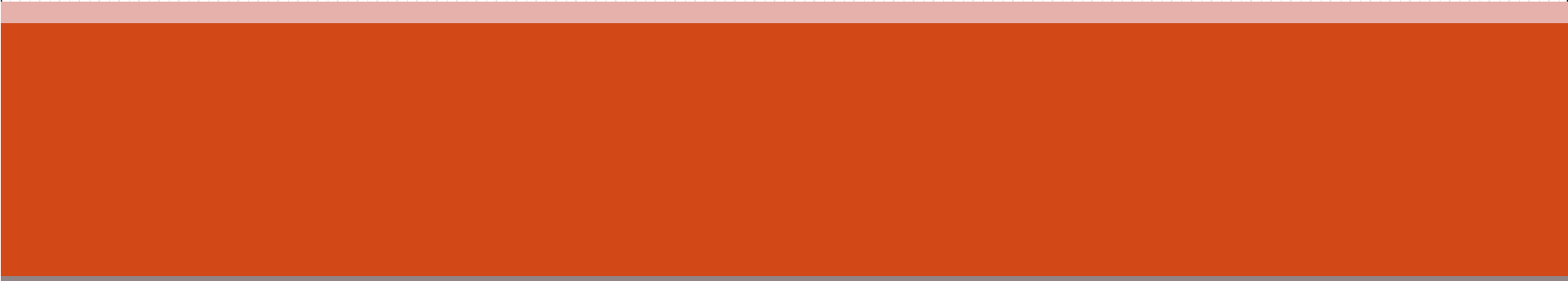
Q_A	$Q_B Q_C$			
	00	01	11	10
0	0	X	X	1
1	X	X	X	1

$$J_C = Q_B$$

For K_C

Q_A	$Q_B Q_C$			
	00	01	11	10
0	X	1	1	X
1	X	X	X	X

$$K_C = 1$$

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THANK YOU !!